

Research article

ASEADOS-SDN-IoT: A novel SDN-IoT network intrusion detection dataset and framework

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ABSTRACT

The convergence of Software-Defined Networking (SDN) and the Internet of Things (IoT) offers a transformative architecture for intelligent, programmable, and scalable network management. However, this integration also exposes new attack surfaces and security challenges that require robust Intrusion Detection Systems (IDS). Progress in IDS research for SDN-IoT environments has been limited by the absence of realistic datasets that capture both IoT traffic diversity and SDN control-plane dynamics. To address this gap, this paper presents ASEADOS-SDN-IoT, a novel, publicly available intrusion detection dataset supported by a fully documented hybrid testbed framework. The proposed framework integrates physical IoT devices and virtual IoT nodes within an ONOS-controlled OpenFlow infrastructure, enabling synchronised collection of packet-flow data and SDN controller telemetry under benign and adversarial conditions. The dataset covers four representative attack categories-Denial-of-Service (DoS), Distributed Denial-of-Service (DDoS), Probe, and Botnet-and captures genuine control-data plane interactions observed during live network operation. ASEADOS-SDN-IoT comprises 457,044 labelled flow instances with 83 statistical features and is publicly released to support reproducible research. The experimental evaluation using machine learning (ML) and deep learning (DL) models demonstrates clear separability between benign and malicious traffic. It confirms the dataset's suitability for training and benchmarking intrusion detection systems. By combining realistic traffic generation, cross-layer visibility, and reproducible design, ASEADOS-SDN-IoT provides a robust benchmark for advancing secure and adaptive SDN-IoT infrastructures.

1. Introduction

The Internet of Things (IoT) has emerged as a transformative paradigm of the 21st century, connecting billions of devices across domains such as healthcare, transportation, manufacturing, energy, and smart cities. These devices, ranging from lightweight sensors and wearables to industrial controllers and smart appliances, generate vast volumes of data that enable real-time monitoring, automation, and intelligent decision-making [1]. Forecasts predicting tens of billions of deployed IoT devices highlight its role as a cornerstone of digital transformation; however, this rapid growth also introduces significant challenges in network management, scalability, and security [2]. IoT ecosystems are inherently heterogeneous, comprising devices built on diverse hardware platforms, operating systems, and software stacks, and employing a wide range of communication protocols [3,4]. At scale, IoT networks must manage millions of dynamic endpoints that frequently join or leave the network or modify their traffic behaviour. At the same time, most IoT devices remain resource-constrained in terms of processing power, memory, and energy. This combination of scale,

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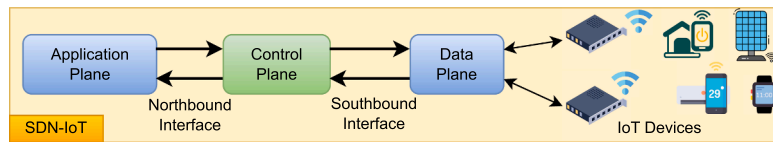


Fig. 1. Overview of SDN-IoT architecture illustrating controller-based coordination of IoT devices.

heterogeneity, dynamicity, and resource limitations makes IoT networks extremely difficult to configure, monitor, and secure using traditional networking approaches [5].

To address the management, scalability, and heterogeneity challenges of large-scale IoT deployments, Software-Defined Networking (SDN) has been widely adopted for its centralised control, programmability, and global network visibility. Prior studies have shown that the convergence of SDN and IoT enables unified management of heterogeneous devices, dynamic traffic reconfiguration, and improved network monitoring capabilities in highly dynamic environments [6–8]. Furthermore, although SDN was not originally designed as a security mechanism, its centralised control and programmability have been shown to facilitate effective intrusion and anomaly detection, as well as automated response, in SDN-IoT ecosystems [7].

This convergence of IoT and SDN, often referred to as the SDN-IoT paradigm, is increasingly seen as a foundation for building secure, scalable, and manageable next-generation networks [9]. With IoT systems embedded in critical infrastructures such as energy grids, healthcare platforms, and industrial control systems, securing SDN-IoT environments has become an urgent priority. Among the many security mechanisms available, Intrusion Detection Systems (IDS) play a central role by monitoring traffic to detect anomalies and malicious behaviors [10,11]. Recent advances in machine learning (ML) and deep learning (DL) have enhanced IDS capabilities; however, their effectiveness fundamentally depends on the availability of realistic and representative datasets, without which IDS models risk poor generalisation and weak real-world performance [12,13].

The evolution of intrusion detection research has been closely tied to the availability of representative datasets that reflect prevailing network architectures and attack landscapes [14]. Early benchmarks, such as KDD99 and NSL-KDD, focused on traditional enterprise networks and standard attack categories, whereas later datasets, such as UNSW-NB15, CICIDS2017, and CSE-CIC-IDS2018, incorporated more diverse and realistic traffic patterns [15–17]. With the emergence of IoT, specialised datasets, including Bot-IoT, IoTID20, and TON-IoT, began to capture device-specific behaviours and IoT-oriented attacks. In contrast, the rise of programmable networks led to SDN-focused datasets, such as InSDN, SDNFlow, and the SDN-DDoS Traffic Dataset, that incorporate controller-level interactions [18,19]. Despite these advances, existing datasets primarily focus on IoT or SDN in isolation and do not adequately capture their convergence. In SDN-IoT environments, prevalent attack classes—including DoS, DDoS, Probe, and Botnet attacks—can simultaneously disrupt IoT devices and SDN control planes, posing serious threats to availability, confidentiality, and integrity [11,20,21]. This gap motivates the need for datasets explicitly tailored to SDN-IoT infrastructures that capture both IoT traffic characteristics and SDN control-plane behaviour [22].

To advance research in this space, we present a novel SDN-IoT intrusion dataset that captures realistic benign and malicious traffic under controlled experimental conditions. The dataset was generated using a hybrid testbed (See Fig. 2) that integrates physical IoT devices, including Raspberry Pi boards and intelligent assistants running real-world applications, with virtual IoT nodes emulated using Mininet, Flask, and Python [23–27]. An ONOS SDN controller managed the programmable infrastructure through Open vSwitch instances, while attack scenarios were orchestrated using Kali Linux and Metasploitable virtual machines [28,29]. The dataset captures both benign SDN-IoT communication and malicious traffic, representing DoS, DDoS, Probe, and Botnet attacks. By incorporating both device-level traffic and SDN-specific controller interactions, the dataset offers a comprehensive resource for training and evaluating IDS models. The contributions of this work are summarised as follows:

- **Hybrid SDN-IoT data collection framework:** We design a hybrid SDN-IoT testbed that integrates physical IoT devices and virtual nodes under centralised SDN control, enabling joint observation of IoT traffic and SDN control-plane behaviour.
- **ASEADOS-SDN-IoT dataset:** We release a publicly available SDN-IoT intrusion detection dataset containing labelled benign traffic and four major attack categories—DoS, DDoS, Probe, and Botnet—captured under realistic operating conditions.
- **Cross-layer SDN-IoT visibility:** Unlike existing datasets, ASEADOS-SDN-IoT captures SDN controller events (e.g., flow installations and packet-in messages) alongside IoT data-plane traffic, enabling cross-layer intrusion analysis.
- **Benchmark validation:** We validate the dataset using ML and DL models, demonstrating its suitability for benchmarking intrusion detection systems in SDN-IoT environments.

In summary, the ASEADOS-SDN-IoT dataset represents the next step in the evolution of IDS benchmarks by focusing explicitly on SDN-IoT environments. It complements existing general-purpose, IoT-specific, and SDN-specific datasets by addressing their convergence, where the scale and heterogeneity of IoT meet the programmability of SDN. By capturing both benign and malicious scenarios in realistic testbed conditions, the dataset provides a foundation for developing intelligent IDS solutions. It contributes to the security of next-generation SDN-IoT networks.

The remainder of this paper is organised as follows: Section 2 presents background on SDN-IoT and associated security challenges; Section 3 reviews related work and existing intrusion detection datasets; Section 4 describes the proposed SDN-IoT architecture; Section 5 details the data generation methodology; Section 6 presents the dataset composition, extracted features, and availability;



Fig. 2. Lab setup for ASEADOS-SDN-IoT dataset generation.

Section 7 reports the experimental evaluation and analysis; Section 8 outlines future research directions; and Section 9 concludes the paper (See Fig. 3)

2. Background

2.1. IoT networks and security challenges

The IoT is a highly heterogeneous ecosystem that integrates a wide variety of devices, ranging from lightweight sensors and actuators to powerful edge computing platforms [30]. These devices are deployed across various domains, including healthcare, smart cities, industrial automation, agriculture, and energy systems. Despite the benefits of connectivity, IoT systems face significant challenges in networking and security [31]. A fundamental issue in IoT networks is device heterogeneity. Devices come from different manufacturers, operate on diverse hardware and software platforms, and support various communication protocols [32]. Lightweight protocols, such as MQTT, CoAP, and AMQP, are widely adopted due to their low overhead. Others utilise proprietary wireless standards, including Zigbee, Bluetooth Low Energy (BLE), or LoRaWAN [33]. This diversity complicates network management, interoperability, and security enforcement.

IoT deployments often involve thousands, or even millions, of devices, each generating data flows with varying frequency and size. Traditional networking approaches, which rely on manual configuration and static control, are not suited for such dynamic environments [5]. Moreover, IoT networks are highly dynamic, with devices frequently joining or leaving, or altering their traffic behaviour depending on operational context. From a security standpoint, IoT devices are often resource-constrained, with limited memory, processing power, and energy. This prevents them from running powerful cryptographic algorithms, intrusion-prevention tools, or frequent software updates [34]. Many devices remain unpatched, leaving them vulnerable to security risks long after deployment. These limitations make IoT devices prime targets for attackers, who can exploit them to gain unauthorised access, steal data, or launch large-scale, coordinated attacks [35].

2.2. SDN in IoT networks

SDN emerged as a paradigm to address the limitations of traditional network management. Its core principle is the separation of the control plane from the data plane, enabling centralised programmability and global visibility [6,36]. The control plane, managed by an SDN controller, is responsible for making routing and policy decisions, while the data plane forwards packets according to flow rules installed by the controller. Communication between the controller and forwarding devices is typically achieved via southbound

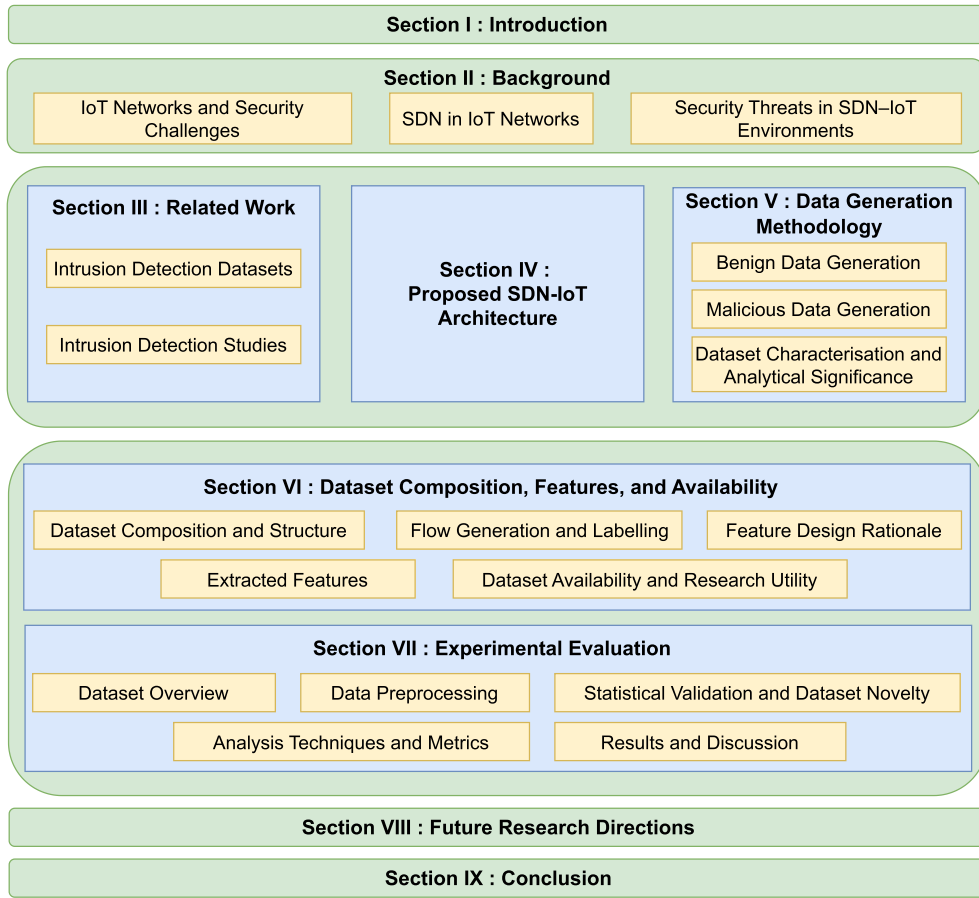


Fig. 3. Overall organisation of the paper, showing the main sections and their corresponding subsections.

protocols such as OpenFlow, while northbound APIs enable interaction with higher-level applications. In the context of IoT, SDN offers several benefits. Centralised policy enforcement allows administrators to configure and monitor heterogeneous devices uniformly [37,38]. Dynamic flow management enables the network to adapt in real time to device churn or traffic fluctuations, ensuring Quality of Service (QoS) for critical applications such as healthcare telemetry or industrial automation. Protocol abstraction simplifies interoperability among IoT devices that use different communication standards, while the controller's global network view enables holistic traffic analysis and anomaly detection [39].

The convergence of IoT and SDN forms the SDN-IoT paradigm, which combines the ubiquity of IoT with the flexibility and manageability of SDN [40,41]. SDN-IoT environments can also be integrated with edge and fog computing, further reducing latency by bringing decision-making closer to data sources [42]. Despite these advantages, SDN-IoT introduces its own challenges, including controller scalability, latency sensitivity, and the risk of a single point of failure. Nevertheless, SDN-IoT is widely recognised as a promising architecture for future intelligent networks [43,44].

2.3. Security threats in SDN-IoT environments

SDN-IoT environments are exposed to a diverse range of security threats due to the convergence of heterogeneous IoT devices and centrally managed SDN control planes [20,43,45]. Among these, four attack categories have been consistently identified in the literature as particularly prevalent and impactful. DoS attacks target resource-constrained IoT devices or services by exhausting computational or communication resources, leading to service degradation or outages. DDoS attacks amplify this impact by coordinating large volumes of malicious traffic, often generated by compromised IoT botnets, which can overwhelm not only end devices but also SDN controllers and controller-switch communication channels [46–48]. Probe attacks, including scanning and reconnaissance activities, enable adversaries to discover network topology, exposed services, and vulnerabilities, and are frequently used as a precursor to more sophisticated intrusions. Botnet attacks are among the most severe threats in SDN-IoT environments, as malware-infected devices can be remotely orchestrated to execute coordinated activities such as DDoS campaigns, malware propagation, and data exfiltration. The Mirai botnet exemplifies the large-scale impact of such attacks on Internet infrastructure [17,49].

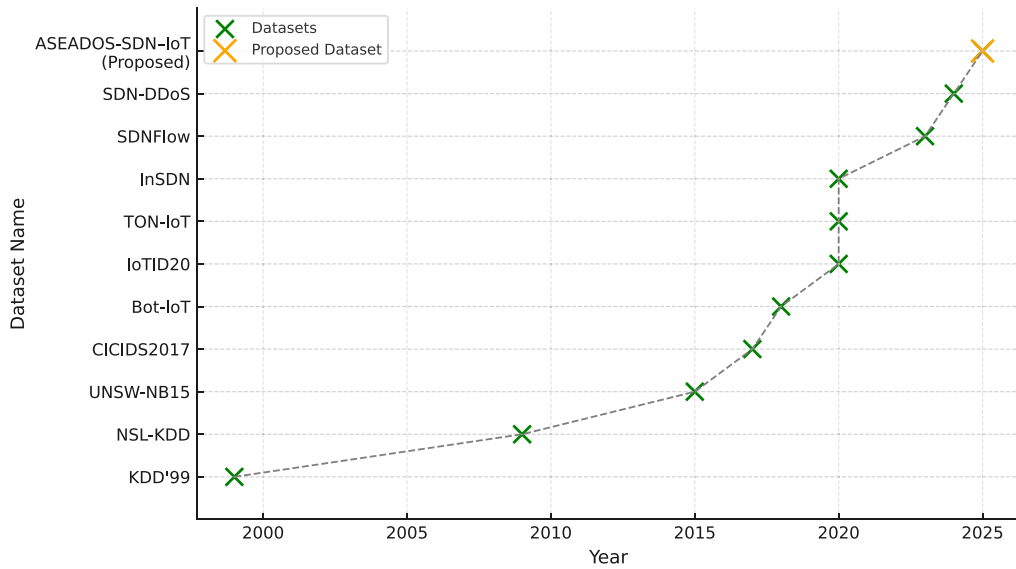


Fig. 4. Chronological overview of key intrusion detection datasets showing the evolution from traditional IDS benchmarks to IoT, SDN and SDN-IoT domains.

Collectively, these attack categories-DoS, DDoS, Probe, and Botnet-pose significant risks to both IoT service availability and SDN control-plane stability [21]. Their prevalence and operational impact motivate their selection as representative threat classes for SDN-IoT intrusion detection research and provide a principled basis for dataset construction and IDS evaluation in realistic SDN-IoT settings.

3. Related work

3.1. Intrusion detection datasets

The progress of IDS research has always depended on publicly available datasets that provide representative traffic for training, validation, and benchmarking. Over time, datasets have evolved from synthetic enterprise traces to more realistic collections reflecting IoT and SDN environments [50]. Below, we review the most influential datasets in depth.

- KDD'99 Dataset:** The KDD Cup 1999 dataset, derived from the DARPA 1998 Intrusion Detection Evaluation Program at MIT Lincoln Laboratory, was the first large-scale benchmark for intrusion detection. Its primary motivation was to provide researchers with a standardised dataset for evaluating IDS techniques in a controlled environment [51]. The methodology involved simulating a U.S. Air Force local area network (LAN) where scripted attacks were combined with benign user activities, such as Telnet sessions, file transfers, and web browsing. Traffic was captured and processed into connection records, each described by 41 features, including basic attributes, content-based attributes, and time-based statistics. The dataset covered four major attack categories: DoS, Probe, R2L, and U2R. KDD'99 became the dominant benchmark for over a decade and played a key role in establishing ML-based IDS research [52]. Later studies highlighted that its synthetic generation and repetitive records limited its ability to accurately reflect evolving real-world traffic, which motivated the development of subsequent datasets [53].
- NSL-KDD Dataset:** The NSL-KDD dataset, introduced by the University of New Brunswick in 2009, was designed as a refined successor to KDD'99, addressing issues of redundancy and class imbalance [54]. It retained 41 features and four main attack categories -DoS, Probe, R2L, and U2R -but applied a sampling strategy that removed duplicate records and balanced the class distribution. It ensured that classifiers were not biased toward frequent patterns and could better detect rare yet critical attacks such as R2L and U2R. By keeping the dataset size manageable, NSL-KDD enabled complete experimentation without the need for down-sampling, improving the consistency of evaluation results. Although no longer actively maintained, it remains a widely cited benchmark and an essential bridge toward more realistic IDS datasets developed in subsequent years [55].
- UNSW-NB15 Dataset:** The UNSW-NB15 dataset, released in 2015 by the Australian Centre for Cyber Security (ACCS) at UNSW Canberra, was designed to capture modern and diverse traffic beyond what earlier datasets offered [56]. It was generated in a controlled cyber range using the IXIA PerfectStorm tool, which produced a hybrid of realistic, benign activities and synthetic attack behaviours. Tcpdump was used to capture approximately 100 GB of raw traffic, which was later processed into 2.54 million records and stored across four CSV files. The dataset contains both normal traffic and nine attack families: Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms. Feature extraction was performed using Argus and Bro-IDS tools, generating 49 labelled features encompassing flow, content, and time-based attributes. Official splits were provided with 175,341

records for training and 82,332 for testing, ensuring consistency across studies. UNSW-NB15 remains a widely used benchmark, offering richer, more contemporary traffic than legacy datasets, and has played a significant role in advancing IDS evaluation [57].

- CICIDS2017 Dataset:** The CICIDS2017 dataset was released by the Canadian Institute for Cybersecurity (CIC) in 2017. It was created to address the limitations of earlier IDS benchmarks and provide a comprehensive, realistic evaluation platform [58]. Traffic was collected over five consecutive days (July 3–7, 2017) in a simulated corporate network with 25 profiled users generating benign activities such as web browsing, VoIP, Email, file transfers, and streaming. Realistic background traffic was generated using the B-Profile system to emulate human interaction patterns. Attack campaigns were executed each day systematically, covering Brute Force (FTP/SSH), DoS and DDoS (Slowloris, LOIC, Hulk), Web Attacks (XSS, SQL Injection), Infiltration (Metasploit, Dropbox-based), Heartbleed, Port Scans, and Botnet traffic. The dataset comprises over 80 flow features extracted using CICFlowMeter, along with raw PCAPs and labelled CSVs, totalling more than 50 GB of traffic. Its richness lies in complete network configuration, attack diversity, and realistic user behaviour, making CICIDS2017 one of the most widely used modern IDS datasets for ML and DL evaluation [59,60].
- CSE-CIC-IDS2018 Dataset:** The CSE-CIC-IDS2018 dataset was released through a collaboration between the Communications Security Establishment (CSE) of Canada and the CIC as an extension of CICIDS2017 [58]. Its goal was to provide a larger and more diverse dataset that better reflects the characteristics of enterprise networks. The testbed was deployed on AWS and consisted of five subnets, featuring over 400 victim machines and 30 servers. Additionally, more than 50 machines were used to launch attacks. Traffic was generated using profile-based methods: B-profiles modelled realistic user activity over standard protocols, and M-profiles defined reproducible attack scenarios. Over 10 days, 7 attack categories were executed: Brute Force, DoS, DDoS, Web attacks, Heartbleed, Infiltration, and Botnet. The dataset comprises raw PCAPs, host logs, and labelled flows, featuring over 80 statistical features extracted using CICFlowMeter-V3. CSE-CIC-IDS2018 remains a comprehensive benchmark widely used in intrusion detection research, particularly for evaluating ML and DL [59,61].
- Bot-IoT Dataset:** The Bot-IoT dataset, developed in 2018 at the UNSW Canberra Cyber Range Lab, was created to address the growing threat of IoT botnets and the lack of IoT-specific benchmarks. A realistic network testbed was designed to generate both normal IoT traffic and malicious traffic, including DoS, DDoS, service scans, keylogging, and data exfiltration [62,63]. Attacks were further categorised by protocol type, while benign traffic captured standard IoT services. The dataset provides source files in multiple formats, including raw PCAPs, Argus files, and CSV feature sets. Its scale is significant: the PCAPs total 69.3 GB and contain over 72 million records, while the extracted flow traffic in CSV format is 16.7 GB. To ease usability, a 5% subset was also released, containing approximately 3 million labelled records. Bot-IoT rapidly became a widely adopted benchmark due to its large scale, realistic IoT focus, and suitability for training and evaluating ML/DL-based intrusion detection models [64,65].
- IoTID20 Dataset:** The IoTID20 dataset, released in 2020, was developed to provide a realistic benchmark for intrusion detection in IoT environments, where lightweight protocols and constrained devices create unique security challenges [66]. Before its introduction, most IDS datasets either focused on traditional enterprise traffic or used synthetic IoT traces, leaving a gap for device-centric data. IoTID20 was generated in a smart home testbed that included two victim devices—an SKT NUGU smart speaker and an EZVIZ Wi-Fi camera—connected alongside laptops, tablets, and smartphones. Benign traffic was interlaced with malicious scenarios, including DoS attacks, Nmap reconnaissance, and Mirai botnet activity, the latter of which was manipulated to emulate IoT-originated flows. A total of 42 PCAPs were captured in wireless monitor mode and processed with CICFlowMeter to extract 83 flow features and three label dimensions (binary, category, and subcategory). By focusing on device-level communication, IoTID20 captures the constraints of IoT ecosystems and remains a crucial benchmark for evaluating intrusion detection in resource-constrained environments [67,68].
- TON_IoT Dataset:** The TON_IoT dataset, introduced in 2020 by UNSW Canberra, was developed to address a key limitation of earlier IDS benchmarks: the lack of heterogeneous data that captures the complexity of IoT and Industrial IoT (IIoT) environments [69]. While previous datasets mainly focused on network flows, TON_IoT integrates three complementary modalities—telemetry from over ten IoT/IIoT sensors (e.g., weather, Modbus, GPS), operating system audit logs from Windows and Linux hosts, and network traffic captured using Zeek. Built on a realistic Industry 4.0 testbed spanning IoT, Fog/Edge, and Cloud layers, it records benign activities alongside diverse attacks, including DoS, DDoS, ransomware, data exfiltration, and web-based exploits. The dataset is provided in raw PCAPs, processed CSVs with labelled features, and train/test-ready subsets, all supported by ground-truth event logs. By bridging device-level, system-level, and network-level perspectives, TON_IoT fills a significant research gap and has become a cornerstone for ongoing IDS and anomaly detection studies in smart homes, smart cities, and IIoT [65,68].
- CICIoT2023 Dataset:** The CICIoT2023 dataset, introduced by the CIC, represents one of the most comprehensive benchmarks for intrusion detection in IoT environments [70]. Built on a large-scale smart home testbed with 105 IoT devices, it captures both benign traffic and 34 modern attack types, including DDoS, DoS, reconnaissance, brute-force, spoofing, Mirai variants, and web-based exploits. The dataset comprises over 14 million records, featuring 47 flow-based attributes extracted using CICFlowMeter 4.0, including packet rates, header lengths, inter-arrival times, and protocol usage. Attacks were generated using realistic tools such as Nmap, Hping3, and Metasploit to ensure fidelity to real-world adversarial techniques. Beyond scale and diversity, CICIoT2023 addresses prior limitations in IoT security datasets by providing balanced labelling for both binary and multi-class classification, enabling evaluation across anomaly detection, supervised learning, and time-series modelling tasks [17,71]. Its size, realism, and protocol heterogeneity make it a valuable foundation for ML, DL, and explainable AI research, while also highlighting practical challenges, such as class imbalance and limited device diversity, that motivate future work on dataset evolution and IDS development.
- InSDN Dataset:** The InSDN dataset, released in 2020 by researchers at University College Dublin, was developed to capture the distinctive traffic behaviours and vulnerabilities of SDN environments [28]. Built on a virtual testbed, the framework incorporated

Table 1
Comparison of major IDS datasets.

Dataset	Year / Origin	Scale	Attack Categories	Features	Format	Domain
KDD'99	1999, MIT Lincoln Lab	5M records (~800MB)	DoS, U2R	Probe, R2L, 41 connection-level features	CSV	Enterprise LAN
NSL-KDD	2009, UNB	~145K records	DoS, U2R	Probe, R2L, 41 features, balanced difficulty levels	ARFF, CSV	Enterprise LAN
UNSW-NB15	2015, UNSW Canberra	2.54M records (100 GB raw)	9 families: DoS, Exploits, Fuzzers, Reconnaissance, etc.	49 flow, content, and time-based features	PCAP, CSV	Enterprise / Hybrid
CICIDS-2017	2017, CIC	Over 2.8 million records (50 GB traffic)	Brute Force, DoS, DDoS, Web Attacks, Infiltration, Heartbleed, Botnet	80+ flow features via CICFlowMeter	PCAP, CSV	Enterprise
CSE-CIC-IDS2018	2018, CIC + CSE	approximately 16 million records	7 categories: DoS, DDoS, Brute Force, Web, Heartbleed, Infiltration, Botnet	80+ statistical flow features, host logs	PCAP, CSV, Logs	Enterprise
Bot-IoT	2018, UNSW Canberra	72M flows (69.3 GB / 5% extracted)	DoS, DDoS, Scan, Keylogging, Exfiltration	40+ features (flows, Argus stats)	PCAP, Argus, CSV	IoT
IoTID20	2020, OTU Canada	625,783 records	DoS, Scan, Botnet	Mirai, 83 features + 3 label dimensions	PCAP, CSV	IoT (smart home)
TON_IoT	2020, UNSW Canberra	257,673 records	DoS, DDoS, Ransomware, Exfiltration, Web attacks	contains 43 features	PCAP, CSV, Logs	IoT / IIoT
CICIoT-2023	2023, CIC	14M+ records	DDoS, DoS, Recon, Brute Force, Spoofing, Mirai, Web-based attacks (34 attacks)	47 flow-based attributes	PCAP, CSV	IoT
InSDN	2020, UCD Ireland	343K instances	DoS, DDoS, Botnet, Recon, SQLi, XSS, Brute Force	80+ features via CICFlowMeter	Flow PCAP, CSV	SDN
SDNFlow	2023, INICTEL-UNI	662K records (221K normal, 441K attack)	37 features	37 OpenFlow switch statistics	CSV	SDN
SDN-DDoS Traffic	2024, Tecnológico de Monterrey	The total size of the dataset is 388 GB.	Diverse DDoS scenarios under two topologies	Flow-level features (delay, jitter, loss, rates)	PCAP, CSV	SDN

the ONOS controller, Mininet with Open vSwitch for topology emulation, and additional servers for generating both benign services and malicious activity. Normal traffic reflected realistic user behaviour across categories such as HTTPS, DNS, SSH, Email, video streaming, voice-over-IP, and file transfers. They systematically executed malicious scenarios, spanning over ten attack families, including DoS and DDoS floods (LOIC, Hping3, HULK, Slowloris), brute-force login attempts, botnet activity (ARES), reconnaissance using tools such as Nmap and Metasploit, and application-layer exploits including SQL injection and cross-site scripting. The dataset comprises more than 343,000 labelled instances, released in both raw PCAP format and processed CSV feature sets extracted using CICFlowMeter. By combining control-plane and data-plane communications with application-level traffic, InSDN provides one of the first comprehensive benchmarks for evaluating IDS and anomaly detection techniques in programmable networks [19,72,73].

- **SDNFlow Dataset:** The SDNFlow dataset was created to address the scarcity of intrusion-detection benchmarks tailored for SDN, particularly those that capture OpenFlow traffic characteristics [29]. Unlike earlier SDN datasets, which primarily relied on packet captures, SDNFlow is generated exclusively from OpenFlow switch statistics collected and processed by an SDN application, thereby reflecting the intrinsic behaviour of programmable networks. The dataset comprises 662,828 records, of which 221,564 correspond to benign traffic and 441,264 to attack scenarios, enabling robust evaluation of binary classification approaches. Each record is described by 37 parameters grouped into five categories: flow identifiers (e.g., flow ID, protocol, source/destination IP addresses and ports, flow duration), packet-based features, byte-based features, flow-timer statistics, and cumulative metrics that aggregate activity across IPs and connections. This design emphasises scalability and fine-grained monitoring within SDN infrastructures. By providing flow-level data directly from the control-data plane interface, SDNFlow offers a distinctive resource for developing and benchmarking ML/DL-based intrusion detection systems, advancing both attack detection and traffic analysis in SDN environments [74].
- **SDN-DDoS Traffic Dataset:** The SDN-DDoS Traffic Dataset, released in 2024, provides a large-scale benchmark for studying DDoS attacks in SDN environments [75]. It was generated using two network topologies, each comprising 12 OpenFlow switches and 24 hosts, orchestrated by a Ryu controller, which enables the simulation of realistic control-plane and data-plane interactions. The

Table 2

Comparison between the proposed ASEADOS-SDN-IoT dataset and publicly available IoT/SDN-IoT datasets.

Dataset	Year	Realistic Traffic	Label	Attack Diversity	Type of Network	No. of Attributes	Format	Network Environment
Bot-IoT	2018	Yes	Yes	Yes	IoT Network	40 +	Packet, Flow	IoT testbed (UNSW)
IoTID20	2020	Yes	Yes	Moderate	IoT Network	83	Packet, Flow	Smart Home IoT
TON-IoT	2020	Yes	Yes	Yes	IoT / IIoT	40 +	Logs, Telemetry	Multi-source IoT
CICIoT2023	2023	Yes	Yes	Yes	IoT Network	47	Packet, Flow	smart home (105 devices)
InSDN	2020	Yes	Yes	Yes	SDN Network	~80	Packet, Flow	Virtual SDN Testbed
SDNFlow	2023	Yes	Yes	Limited	SDN Network	37	Flow	OpenFlow Stats
SDN-DDoS	2024	Yes	Yes	No	SDN Network	~45	Flow	Ryu-based SDN
SDN-IoT	2025	Partial	Yes	Yes	SDN-IoT (Synthetic)	~80	CSV	Combined datasets
Proposed	2025	Yes	Yes	Yes	SDN-IoT Network	83	Packet, Flow	SDN-IoT Testbed

dataset consists of 1,048,575 labelled records, encompassing both benign flows and diverse DDoS scenarios. Flow-level features, including packet and byte counts, duration, transmission rates, delay, jitter, and packet loss, are described for each entry. By combining scale, feature richness, and SDN-specific traffic, this dataset fills a critical gap for evaluating anomaly detection and mitigation strategies in programmable networks [76]. It is particularly valuable for training and validating ML and DL models designed for the real-time detection of large-scale flooding attacks in SDN infrastructures.

Collectively, these datasets demonstrate the evolution of IDS benchmarks, progressing from early synthetic datasets to modern, domain-specific collections for IoT and SDN. Each has played an essential role in advancing IDS research and has addressed the needs of its time. However, the convergence of IoT and SDN remains underrepresented, as most available datasets focus on either IoT or SDN individually. This gap motivates the development of new datasets that explicitly capture SDN-IoT environments, enabling research on intrusion detection in this emerging paradigm.

3.2. Intrusion detection studies

Intrusion detection has long been a central component in protecting networked systems, with its early developments rooted in traditional enterprise networks. Signature-based IDS provided reliable detection of known threats but proved ineffective against zero-day and evolving attacks. Anomaly-based methods emerged as an alternative, capturing deviations from expected behaviour through statistical and later ML approaches. Mondragon et al. conducted a systematic literature review on IDS datasets, highlighting that while early benchmarks, such as DARPA 1998 and KDD'99, shaped anomaly detection research, their synthetic nature and limited scope restricted realism [77]. In another study by Ji et al., this critique was reinforced, noting that even improved datasets, such as NSL-KDD and UNSW-NB15, failed to capture the heterogeneity of modern traffic [78]. A meta-analysis by Maseer et al. described this phenomenon as benchmark-driven progress, where accuracy improvements largely reflected dataset-specific tuning rather than resilience in real-world conditions [79]. Collectively, these studies underscore that from the outset, IDS research has been dataset-driven, with methodological advances constrained by unrealistic evaluation foundations.

A study by Abdulkareem et al. extended this discussion to the IoT context, arguing that the heterogeneity of devices and protocols made machine and DL indispensable for intrusion detection [80]. However, their survey revealed a key weakness: learning-based methods performed well on device-centric datasets but failed to capture the broader context of IoT networking, resulting in models that lacked real-world applicability. A study by Racherla et al. introduced Deep-IDS, a DL-based real-time IDS for IoT nodes, which achieved strong detection accuracy but was computationally too heavy for constrained devices [81]. Similarly, Al-Shurbaji et al. reviewed DL-based IDS for IoT botnet detection, reporting promising results but warning of vulnerability to dataset bias and poor adaptability to evolving botnets [82]. Another investigation by Rakine et al. surveyed ML/DL-based IoT IDS approaches and concluded that although DL models are effective in controlled evaluations, their complexity restricts practical deployment at the edge [83]. Ozdogan et al. presented a comprehensive comparative assessment of ML algorithms for IoT IDS, highlighting high false-positive rates and limited generalisation across datasets [84]. Taken together, these IoT studies demonstrate methodological sophistication but also reveal persistent barriers: device-centric datasets, resource constraints, and insufficient validation under adversarial or evolving conditions.

Janabi et al. surveyed IDS approaches tailored for SDN, emphasising how OpenFlow statistics and flow-level features serve as the basis for anomaly detection [85]. They reported strong results for both ML and DL techniques but warned that these models often neglected scalability and encryption challenges. In healthcare contexts, Asif et al. introduced a hybrid SVM-KNN IDS for SDN infrastructures, achieving improved latency and detection accuracy in controlled testbeds but without validation at scale [86]. In parallel, Arevalo-Herrera et al. surveyed vulnerabilities targeting SDN controllers, noting that most existing studies still rely on outdated controllers (e.g., Floodlight, Ryu) and generic datasets not tailored to SDN dynamics [87]. They highlighted a significant limitation in the field: the scarcity of publicly available datasets that reflect controller-specific attacks, which limits reproducibility and the development of robust ML defences. Collectively, these studies demonstrate that while SDN-based IDS frameworks benefit from ML and programmability, they remain limited by datasets such as InSDN, SDNFlow, and SDN-DDoS. As a result, reported gains often reflect dataset-specific optimisation rather than resilience against adversarial and heterogeneous SDN-IoT environments.

In addition, Sarika et al. developed a novel IDS for SDN-based IoT networks that integrates ML and DL techniques, combining classifiers such as XGBoost, decision trees, and logistic regression with recurrent models including LSTM, RNN, and GRU [13]. Their framework incorporated an optimised preprocessing pipeline, achieving very high accuracy. LSTM and XGBoost surpassed traditional

Table 3

Summary of related works: Contributions and limitations.

Paper	Year	Focus	Contribution	Limitations
Mondragon et al. [77]	2025	Systematic review of IDS datasets	Analyzed early and modern datasets (DARPA, KDD'99) for anomaly detection research	Synthetic, outdated, limited scope; restricted realism
Ji et al. [78]	2024	Review of IDS in CPS networks	Highlighted dataset evolution (NSL-KDD, UNSW-NB15)	Still failed to capture traffic heterogeneity in modern environments
Maseer et al. [79]	2024	Meta-analysis of anomaly IDS	Described benchmark-driven progress in IDS research	Accuracy often dataset-specific, poor real-world resilience
Abdulkareem et al. [80]	2024	IoT and non-IoT IDS survey	Showed ML/DL as indispensable for heterogeneous IoT	Device-centric datasets, lack broader IoT networking context
Racherla et al. [81]	2024	Deep-IDS for IoT	Proposed DL-based real-time IDS for IoT nodes	High computational cost, unsuitable for constrained devices
Al-Shurbaji et al. [82]	2025	IoT botnet detection review	Reviewed DL-based IDS for IoT botnets	Dataset bias, poor adaptability to evolving botnets
Rakine et al. [83]	2024	IoT IDS survey	Surveyed ML/DL IoT IDS approaches	Complexity of DL models prevents edge deployment
Ozdogan [84]	2024	ML algorithms in IoT IDS	Comparative analysis of ML algorithms	High false positives, limited cross-dataset generalization
Janabi et al. [85]	2024	SDN IDS survey	Surveyed SDN IDS using OpenFlow and flow features	Neglected scalability, encrypted traffic, adversarial handling
Asif et al. [86]	2025	Healthcare SDN IDS	Hybrid SVM-KNN IDS improved latency and detection accuracy	Validated only in controlled testbeds, no large-scale validation
Arevalo-Herrera et al. [87]	2025	SDN controller vulnerabilities	Surveyed attack types, detection mechanisms, datasets	Relied on outdated controllers and generic datasets, few public datasets for controller attacks
Sarika & Dass [13]	2025	Novel IDS for SDN-IoT	Developed hybrid ML-DL IDS with preprocessing, high accuracy on NSL-KDD, CIC-IDS2017	Legacy datasets, no IoT-SDN realism or control-plane data
Dhirar & Hamad [12]	2025	SDN-IoT dataset	Combined InSDN, BoT-IoT, ToN-IoT; evaluated with CNN, LSTM, RNN, DNN	Merged dataset, not real testbed; lack of generalization across datasets

baselines on NSL-KDD and CIC-IDS2017. However, the evaluation was restricted to these legacy datasets, which omit SDN control-plane dynamics and IoT-specific traffic, limiting the model's relevance to real-world programmable SDN-IoT environments. More recently, Dhirar and Hamad presented a novel SDN-IoT IDS dataset by combining existing sources such as InSDN, BoT-IoT, and ToN-IoT, and evaluated it using CNN, LSTM, RNN, and DNN models [12]. While they reported high accuracies (above 96% for LSTM and DNN), the dataset does not originate from a real SDN-IoT network. This reliance on merged datasets limits its ability to capture genuine control-plane data interactions with IoT, reducing its value for benchmarking intrusion detection in converged environments. Across these contributions, the consensus is clear: SDN-IoT IDS research is promising; however, its progress remains fundamentally constrained by the lack of comprehensive datasets capturing both IoT device traffic and SDN control-plane interactions.

In summary, prior research shows a progression from traditional enterprise IDS to IoT- and SDN-specific solutions, and, more recently, to SDN-IoT convergence. However, across all domains, evaluations still rely on synthetic, outdated, or limited datasets, leading to models that perform well in controlled experiments but fail to generalise in real-world conditions. This persistent reliance on incomplete benchmarks has left SDN-IoT IDS research fragmented, with no widely adopted dataset that captures both IoT device traffic and SDN control-plane interactions. Our work addresses this gap by introducing a comprehensive SDN-IoT dataset to support reproducible evaluation and enable the development of robust, scalable, and deployment-ready IDS solutions. Given these limitations, the next section introduces the proposed SDN-IoT architecture that serves as the foundation for generating our novel dataset. [Table 4](#) summarises a comparison between ASEADOS-SDN-IoT and closely related IoT, SDN and SDN-IoT intrusion detection datasets, highlighting key design and coverage differences.

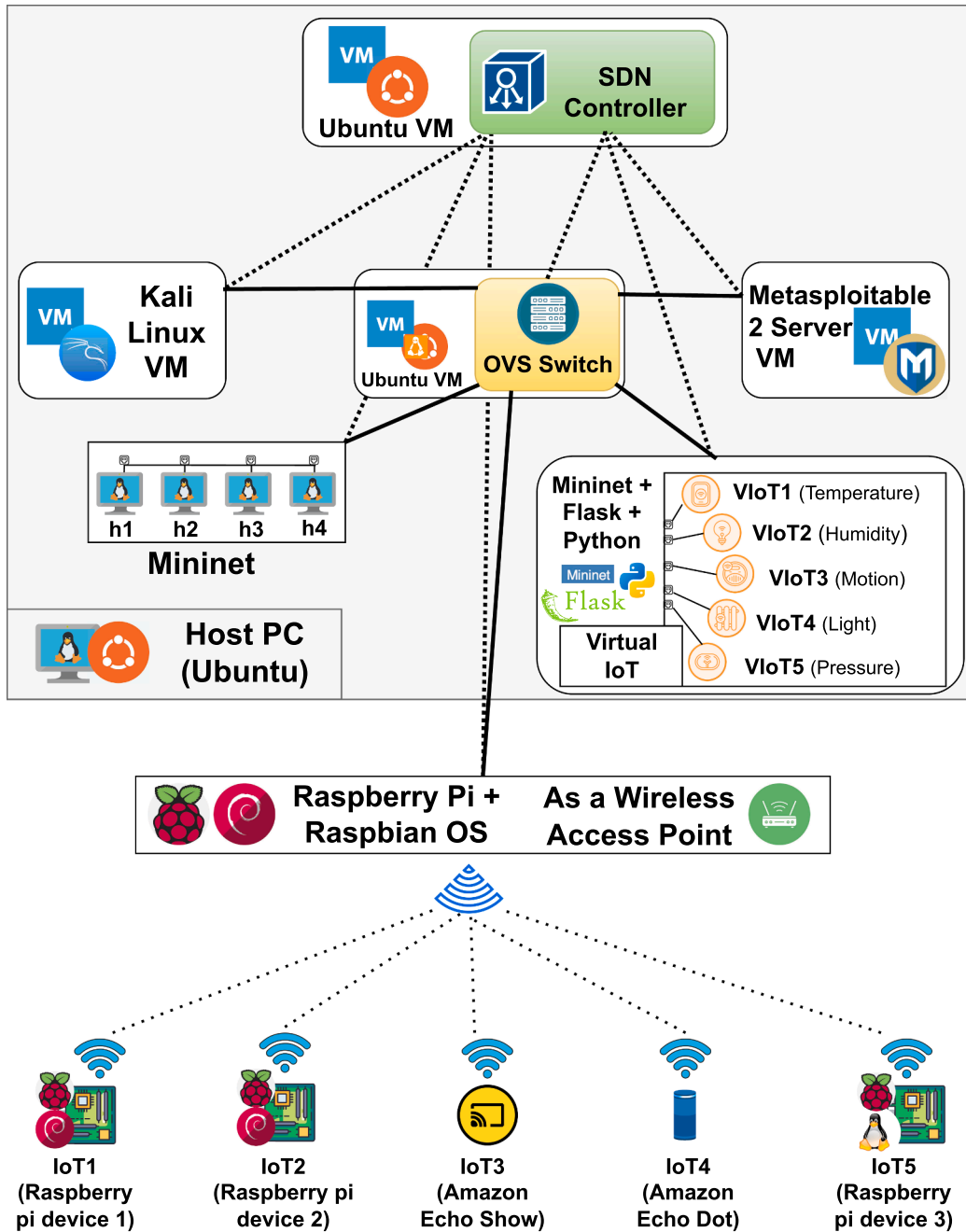


Fig. 5. High-level SDN-IoT architecture.

Table 4

Comparison of ASEADOS-SDN-IoT with closely related IoT, SDN and SDN-IoT datasets.

Dataset	Physical IoT	SDN Control-Plane	Traffic Generation	Attack Coverage
Bot-IoT	×	×	Virtual IoT	DoS, DDoS, Botnet
TON-IoT	✓	×	Hybrid (Physical + Virtual)	Multiple
SDNFlow	×	✓	Virtual SDN	DoS, Probe
SDN-DDoS	×	✓	Virtual SDN	DDoS
InSDN	×	✓	Virtual SDN	Multiple
ASEADOS-SDN-IoT	✓	✓	Hybrid (Physical + Virtual)	DoS, DDoS, Probe, Botnet

4. Proposed SDN-IoT architecture

The proposed SDN-IoT architecture is designed to provide a realistic, controllable environment that models the behaviour of modern IoT systems integrated with SDN networks. The main goal of this architecture is to generate realistic, benign, and malicious traffic while capturing the key characteristics of SDN-IoT interactions. The architecture combines both physical and virtual components to reflect the heterogeneous nature of IoT networks and the centralised control features of SDN. This hybrid setup enables a comprehensive study of network traffic behaviour under various operating conditions and attack scenarios, supporting research on intrusion detection and network management in programmable IoT environments.

The testbed was constructed on a host system running Ubuntu 20.04, with four primary virtual machines (VMs) created in VMware Workstation to represent different network entities and operational roles. Each VM operates as a distinct functional node contributing to the SDN-IoT ecosystem. The first VM runs Kali Linux, serving as the attacker node responsible for launching various attacks, including DoS and DDoS attacks, as well as scanning and probing activities. The second VM runs Ubuntu 18.04 and hosts the ONOS (Open Network Operating System) controller, which represents the SDN control plane [88]. The third VM is based on Metasploitable2 and acts as a vulnerable server that exposes insecure services to emulate compromised IoT endpoints. The fourth VM, also running Ubuntu 18.04, hosts the Open vSwitch (OVS) instance, which forms the programmable data plane and serves as the central interconnection point for all network segments [28,89].

The proposed architecture consists of a single OVS instance, configured with five logical bridges, each assigned to a specific network domain. The first bridge (br1) connects to the Kali Linux attacker VM, enabling the generation of external attack traffic. The second bridge (br2) connects to the Metasploitable2 VM, representing the vulnerable target host. The third bridge (S1) connects to Mininet virtual hosts that emulate legitimate users and background services that generate regular network traffic [90]. The fourth bridge (S2) interfaces with the physical IoT environment, where a Raspberry Pi board running Raspbian OS functions as a wireless access point. This access point connects real IoT devices—including Raspberry Pi boards, Amazon Echo Show, and Echo Dot—to the SDN-controlled network through a bridged connection [91,92]. Finally, the fifth bridge (S3) supports a virtual IoT environment implemented with Mininet, Flask, and Python that emulates sensor nodes generating continuous telemetry, including temperature, humidity, pressure, light intensity, and motion readings. All bridges are centrally controlled by the ONOS controller, which communicates with the OVS switch using the OpenFlow protocol. The controller manages flow rules, collects network statistics, and maintains a global view of the topology, providing the programmability and centralised management that define SDN architectures. The control plane (ONOS) and the data plane (OVS) are deployed on separate virtual machines, closely reflecting real-world SDN deployments and allowing precise observation of their interactions [88]. This design enables the monitoring of control-plane operations such as flow installations, updates, and topology changes—factors that are rarely included in conventional IoT datasets but are critical for understanding how SDN controllers respond to dynamic IoT traffic and attack events.

The architecture adopts a hybrid connectivity model that combines Layer 2 (switching) and Layer 3 (routing) functions. While OVS bridges traditionally operate at Layer 2, in this design, they are integrated with the Linux kernel's Layer 3 routing functionality [93]. This integration allows communication between multiple subnets while preserving SDN control and visibility. The hybrid configuration closely resembles real IoT networks, where devices often operate in isolated subnets for security and scalability, yet depend on a centralised controller for dynamic routing and policy enforcement. The ability to observe routing and switching behaviours simultaneously offers deeper insight into how SDN-based control affects packet forwarding, latency, and throughput in complex IoT environments.

The virtual IoT layer, hosted on bridge S3, generates traffic that reflects typical IoT communication patterns. Each virtual sensor, implemented using Flask and Python, periodically sends data representing environmental readings. These devices emulate real-world IoT nodes by generating continuous, low-volume, and short-duration flows that mimic telemetry updates. The physical IoT layer, connected via bridge S2, introduces real wireless network characteristics into the testbed. The Raspberry Pi-based access point bridges wireless and wired domains, allowing traffic from real IoT devices to flow through the programmable SDN network. This combination of virtual and physical IoT components creates a heterogeneous communication environment that more accurately reflects real-world deployments, where diverse device types and communication protocols coexist.

The overall topology supports both benign and malicious traffic generation. The benign traffic originates from Mininet hosts and IoT devices, which engage in normal network activities such as web browsing (HTTP/HTTPS), file transfers (FTP), DNS queries, and SSH sessions. Meanwhile, malicious traffic is introduced through the Kali Linux attacker and the compromised Metasploitable2 host. The attack campaigns include flooding-based DoS/DDoS attacks, reconnaissance scans, and botnet-like command-and-control behaviours. The simultaneous presence of legitimate and malicious traffic creates a complex network state that challenges intrusion detection systems, enabling researchers to evaluate their accuracy and robustness under realistic mixed-traffic conditions.

Figs. 5 and 6 illustrate the physical and logical topology of the proposed SDN-IoT testbed. The ONOS controller serves as the system's centralised brain, governing the forwarding behaviour of the OVS switches and managing flows across the five bridges. Both virtual and physical IoT devices interact through this unified programmable framework, allowing the controller to observe and manage communication patterns across all domains. The hierarchical structure, comprising the control plane, data plane, and IoT edge layer, captures the essential relationships and interactions that define SDN-IoT systems. The following process outlines the steps to set up the proposed testbed.

- Install and configure **Open vSwitch (OVS)** and **Mininet** on the same Ubuntu 18.04 virtual machine to serve as the data-plane environment.

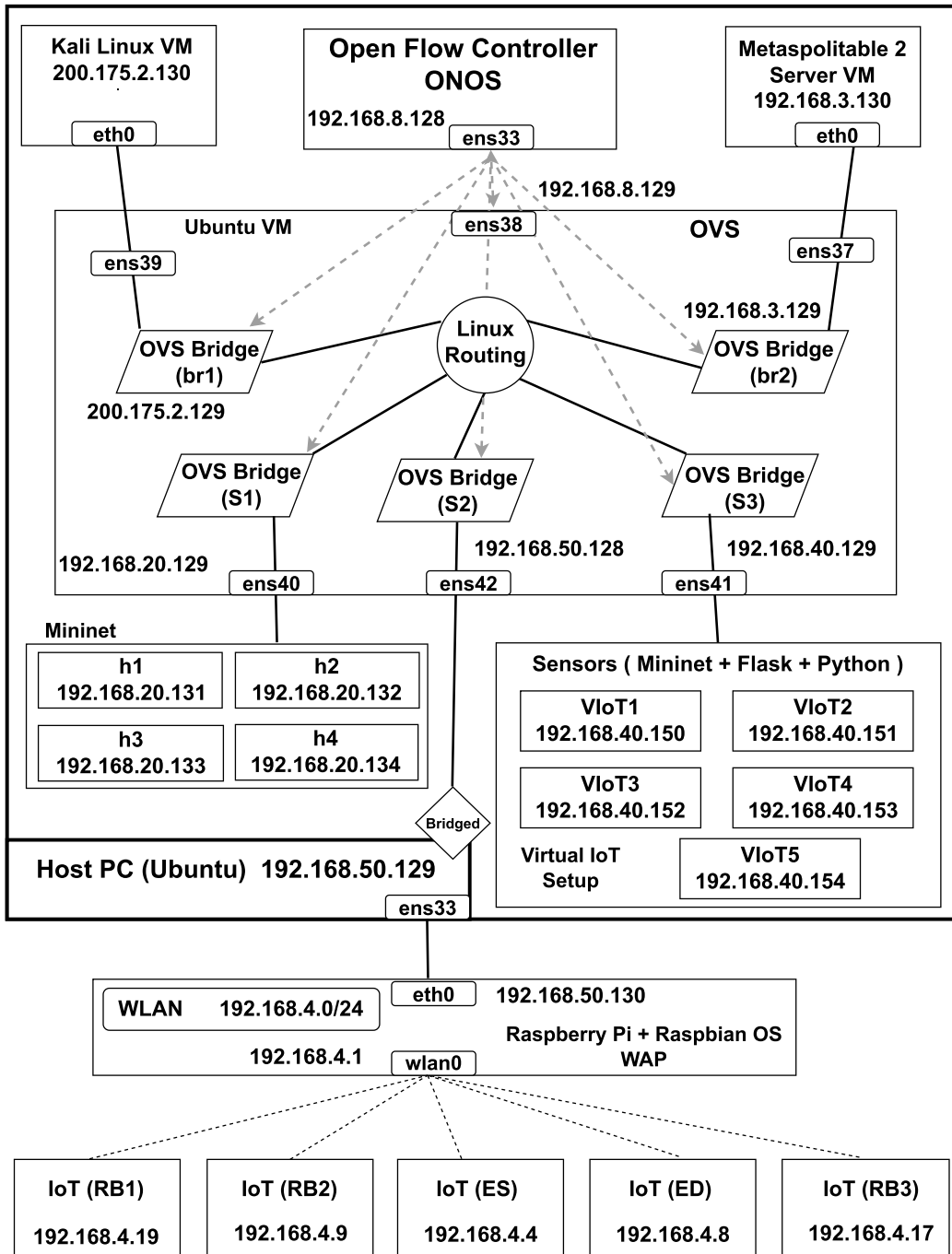


Fig. 6. Low-level SDN-IoT architecture.

- Create five logical OVS bridges (br1, br2, S1, S2, and S3) within a single OVS instance to segment the network according to its functional domains.
- Configure the OVS VM with multiple network interfaces (ens37, ens38, ens39, ens40, ens41, and ens42) to represent different subnets and bridge connections. Each interface is mapped to a corresponding OVS bridge.
- Assign br1 (IP: 200.175.2.129) to connect with the Kali Linux VM (200.175.2.130) for generating external attack traffic.
- Assign br2 (IP: 192.168.3.129) to connect with the Metasploitable2 VM (192.168.3.130), representing a vulnerable internal server.

- Configure S1 (IP: 192.168.20.129) as the Mininet bridge. Attach four virtual hosts (h1–h4) with IP addresses ranging from 192.168.20.131 to 192.168.20.134. These hosts emulate normal network users and generate legitimate background traffic.
- Create S2 (IP: 192.168.50.128) to interconnect with the Raspberry Pi board functioning as a wireless access point. The Raspberry Pi (IP: 192.168.50.129) serves as a bridge, routing traffic between real IoT devices (e.g., Raspberry Pi boards, Amazon Echo Show, and Echo Dot) and the OVS switch.
- Configure S3 (IP: 192.168.40.129) to host the virtual IoT environment implemented with Mininet, Flask, and Python. Deploy five virtual IoT devices (VIoT1–VIoT5) with IP addresses from 192.168.40.150 to 192.168.40.154, each emulating specific sensors such as temperature, humidity, pressure, light, and motion.
- Connect the ONOS controller (Ubuntu 18.04 VM, 192.168.8.128) to the OVS data-plane VM (192.168.8.129) through a dedicated management network. The controller communicates with all OVS bridges via **OpenFlow** (port 6653), enabling control-data plane integration and centralised flow management across the SDN-IoT testbed.
- Enable Linux routing and IP forwarding on the OVS VM to allow communication between subnets through the hybrid Layer 2/Layer 3 model.
- Validate connectivity by performing end-to-end ping tests between the attacker VM, vulnerable server, Mininet hosts, virtual IoT devices, and real IoT devices across all subnets.

In summary, the proposed SDN-IoT architecture offers a comprehensive and realistic platform for studying the interplay between programmable network control and IoT communication. Its integration of physical and virtual devices, centralised SDN management, and hybrid Layer 2/Layer 3 connectivity mirrors the operational dynamics of real IoT networks. This architecture enables the generation of representative traffic that includes both normal and malicious behaviours, supporting the development and evaluation of advanced intrusion detection and mitigation strategies in SDN-IoT environments. The complete testbed framework and documentation are publicly available on GitHub,¹ to facilitate further research, extension, and benchmarking by the scientific community.

5. Data generation methodology

The primary goal is to capture the real-world network behaviour of an SDN-IoT environment under both benign and adversarial conditions. Unlike conventional intrusion datasets that limit observation to packet traces, the proposed work integrates multi-layer visibility. That combines IoT application traffic, SDN flow statistics, and SDN controller telemetry. This approach enables the simultaneous observation of the data plane, which reflects observable network traffic patterns, and the control plane, which reveals the network's cognitive response through the ONOS controller's decision-making process. The resulting dataset not only encapsulates the statistical properties of normal and malicious traffic but also documents how the SDN controller adapts, saturates, or reconfigures itself during dynamic security events.

The generation process was conducted within the hybrid testbed described in Section 3, where physical IoT devices, virtual Mininet hosts, and virtual sensors coexisted under centralised ONOS control. Each experiment was designed to emulate a specific operational context, where benign activity represented continuous service interactions. At the same time, malicious traffic introduced temporal perturbations representative of modern SDN-IoT-based cyber threats. Integrating controller-plane behaviour enables the dataset to capture how the SDN controller manages network events in real time-tracking flow installations, packet-in frequencies, and topology updates. Such visibility, rarely present in conventional datasets, is crucial for understanding how the control logic reacts to disturbances and maintains network stability during attacks.

5.1. Benign data generation

Two distinct benign datasets were recorded to represent the steady-state behaviour of the SDN-IoT ecosystem. The first benign dataset captures IoT device communications and host-generated traffic flows that characterise typical operations in a smart network environment. Virtual IoT devices were implemented in Mininet and combined with Flask and Python scripts to emulate real-time sensor telemetry, including temperature, humidity, motion, and light intensity. These devices generated periodic, low-rate UDP and HTTP messages, reflecting common IoT reporting patterns. Simultaneously, Mininet hosts generated high-level application traffic-HTTP/HTTPS requests, DNS lookups, FTP file transfers, and SSH sessions-to reflect background services in user-driven networks. To introduce authentic wireless variation, physical IoT nodes, including Raspberry Pi boards and Amazon Echo devices, were connected through a Raspberry Pi access point that bridged the wireless and wired domains. This hybrid arrangement enabled the dataset to incorporate real-world latency, packet jitter, and signal interference effects that are not replicable in purely virtual environments.

The second benign dataset focuses on SDN control-plane dynamics recorded by the ONOS controller during normal operations. It includes continuous logs of packet-in messages, flow installations, topology updates, and statistical polling exchanges between the controller and the Open vSwitch bridges. These records illustrate ONOS's reactive and proactive control behaviour when managing diverse IoT and host traffic flows. Analysing this telemetry alongside the data-plane traces provides a dual-perspective view of network health-capturing how ONOS allocates rules, adapts to topology changes, and maintains forwarding performance in the absence of adversarial interference. The dual benign sets thus form the empirical foundation for distinguishing legitimate operational fluctuations from malicious anomalies in later analysis.

¹ <https://github.com/tharinduyasarathna/ASEADOS-SDN-IoT-A-Novel-SDN-IoT-Network-Intrusion-Detection-Dataset-and-Framework.git>

Table 5

Configuration of attack generation in the ASEADOS-SDN-IoT testbed, detailing DoS, DDoS, Botnet, and Probe attacks along with corresponding tools, attacker hosts, and victim networks.

Attack Classes	Attack Tools	Attacker Machine	Victim-Network
DoS	Hulk, slowhttptest, torshammer	Kali-Linux: 200.175.2.130	h4: 192.168.20.134 RB2: 192.168.4.9 VioT5:192.168.40.154 MetaS 2: 192.168.3.130
DDoS	Hping3	h1:192.168.20.131 h2:192.168.20.132 RB1:192.168.4.19 RB3:192.168.4.17 VioT1:192.168.40.150 VioT2:192.168.40.151	h4: 192.168.20.134 MetaS 2: 192.168.3.130 RB2:192.168.4.9 VioT5:192.168.40.154
Botnet	BoNeSI	Kali-Linux: 200.175.2.130	h1:192.168.20.131 h2:192.168.20.132 RB1:192.168.4.19 RB3:192.168.4.17 VioT1:192.168.40.150 VioT2:192.168.40.151
Probe	Nmap	Kali-Linux: 200.175.2.130	h1:192.168.20.131 h2:192.168.20.132 h3:192.168.20.133 h4:192.168.20.134 RB1:192.168.4.19 RB2: 192.168.4.9 RB3:192.168.4.17 EchoShow:192.168.4.4 EchoDot:192.168.4.8 VioT1:192.168.40.150 VioT2:192.168.40.151 VioT3:192.168.40.152 VioT4:192.168.40.153 VioT5:192.168.40.154 MetaS 2: 192.168.3.130
	Metasploit framework	Kali-Linux: 200.175.2.130	MetaS 2: 192.168.3.130

5.2. Malicious data generation

To reflect realistic threat conditions, four principal attack categories -DoS, DDoS, Botnet, and Probe -were executed with multiple parameter variations, each representing a distinct adversarial strategy. These attacks were injected into the live network while benign activities persisted, ensuring natural background congestion and authentic cross-traffic interference. Each campaign was logged in detail, with timing, tool configuration, and duration recorded to establish precise ground truth.

- **DoS Attacks:** Denial-of-service campaigns were conducted using tools such as Hulk, slowhttptest, and Torshammer, which collectively emulate both volumetric and low-rate application-layer exhaustion attacks. In the data plane, these attacks were characterised by a sudden escalation in packet rates toward specific targets, leading to buffer saturation, increased retransmission ratios, and sharp fluctuations in flow density. The subtlety of slow-rate attacks, such as slowhttptest, was evident in persistent, long-duration flows with low packet volumes but high connection counts—precisely the kind of deceptive traffic that challenges rule-based detectors. At the control plane, ONOS recorded steep increases in packet-in and flow-mod frequencies, revealing how repeated flow requests can strain control-channel capacity and delay routing decisions (See Fig. 7). This dual-layer observation reveals that even localised application-layer floods can propagate systemic effects across SDN infrastructure, affecting both control responsiveness and IoT service quality.
- **DDoS Attacks:** Distributed Denial-of-Service scenarios were executed using Hping3, coordinating multiple virtual and IoT nodes to launch concurrent TCP-SYN, UDP, and ICMP floods. The objective was to emulate the large-scale coordination typical of IoT botnets targeting SDN-controlled networks. In the data plane, DDoS attacks generate highly synchronised, multi-source flows converging on common victims, resulting in entropy collapse in destination IP and port distributions, as well as measurable throughput inconstancies. ONOS logs revealed sustained packet-in congestion, delayed flow-mod responses, and occasional timeouts during peak attack phases—demonstrating how distributed floods propagate vertically, from edge congestion to centralised controller saturation. These experiments underline a crucial property of SDN-IoT networks: the tight coupling between distributed forwarding stress and centralised decision bottlenecks.

No.	Time	Source	Destination	Protocol	Length	Info
698142	413.241260	192.168.8.129	192.168.8.128	OpenFlow	176	Type: OFPT_PACKET_IN
698143	413.241424	192.168.8.129	192.168.8.128	OpenFlow	176	Type: OFPT_PACKET_IN
698144	413.241425	192.168.8.129	192.168.8.128	OpenFlow	177	Type: OFPT_PACKET_IN
698145	413.241651	192.168.8.129	192.168.8.128	OpenFlow	184	Type: OFPT_PACKET_IN
698147	413.242087	192.168.8.128	192.168.8.129	OpenFlow	174	Type: OFPT_PACKET_OUT
698148	413.242153	192.168.8.128	192.168.8.129	OpenFlow	174	Type: OFPT_PACKET_OUT
698149	413.242155	192.168.8.128	192.168.8.129	OpenFlow	175	Type: OFPT_PACKET_OUT
698150	413.242227	192.168.8.128	192.168.8.129	OpenFlow	174	Type: OFPT_PACKET_OUT
698151	413.242592	192.168.8.129	192.168.8.128	OpenFlow	176	Type: OFPT_PACKET_IN
698153	413.242818	192.168.8.128	192.168.8.129	OpenFlow	174	Type: OFPT_PACKET_OUT
698154	413.242913	192.168.8.129	192.168.8.128	OpenFlow	184	Type: OFPT_PACKET_IN
698155	413.243088	192.168.8.128	192.168.8.129	OpenFlow	174	Type: OFPT_PACKET_OUT


```

> Frame 698144: 177 bytes on wire (1416 bits), 177 bytes captured (1416 bits)
> Linux cooked capture v1
> Internet Protocol Version 4, Src: 192.168.8.129, Dst: 192.168.8.128
> Transmission Control Protocol, Src Port: 42834, Dst Port: 6653, Seq: 27093272, Ack: 25373894, Len: 109
▼ OpenFlow 1.3
  Version: 1.3 (0x04)
  Type: OFPT_PACKET_IN (10)
  Length: 109
  Transaction ID: 0
  Buffer ID: OFP_NO_BUFFER (4294967295)
  Total length: 67
  Reason: OFPR_ACTION (1)
  Table ID: 0
  Cookie: 0x000100008b71d724
> Match
  Pad: 0000
▼ Data
  ▼ Ethernet II, Src: VMware_8e:f9:c6 (00:0c:29:8e:f9:c6), Dst: VMware_df:47:16 (00:0c:29:df:47:16)
    > Destination: VMware_df:47:16 (00:0c:29:df:47:16)
    > Source: VMware_8e:f9:c6 (00:0c:29:8e:f9:c6)
    Type: IPv4 (0x0800)
  ▼ Internet Protocol Version 4, Src: 200.175.2.130, Dst: 192.168.4.17
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 53
    Identification: 0x79a6 (31142)
    > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x3132 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 200.175.2.130
    Destination Address: 192.168.4.17
    > Transmission Control Protocol, Src Port: 39166, Dst Port: 80, Seq: 282, Ack: 487, Len: 1

```

Fig. 7. Wireshark capture illustrating the structural characteristics of OpenFlow 1.3 communication and packet encapsulation, showing interactions between the ONOS controller, Open vSwitch, the Kali Linux attack node, and Raspberry Pi-based IoT devices within the SDN-IoT testbed.

- **Botnet Activity:** Coordinated botnet behaviour was simulated using the BoNeSI (Botnet Simulation Environment) tool, which supports the creation of virtual bots capable of generating C2 (command-and-control) traffic, scanning, and synchronised attack bursts. The configuration enabled a mixture of steady-state beaconing traffic and high-frequency attack surges, representing the alternating dormant and active phases typical of real IoT botnets. In the data plane, BoNeSI produced temporally correlated low-rate flows interspersed with periodic flooding, resulting in rhythmic variations in flow density and duration. At the control plane, ONOS exhibited cyclical packet-in and flow-mod oscillations synchronised with these coordination intervals, demonstrating how persistent low-rate bot activity can impose continuous baseline load on the controller. This experiment provides a crucial insight into SDN-IoT security: botnets do not need to cause volumetric saturation to degrade network performance; sustained, coordinated background chatter can exhaust control resources over time.
- **Probe and Exploitation Attacks:** Reconnaissance and exploitation operations were generated using Nmap and the Metasploit Framework to emulate the full pre-attack and exploitation lifecycle. Nmap scanning encompasses host discovery, port enumeration, and version fingerprinting, resulting in numerous short-duration connections with minimal payload and high port variability. Once vulnerable endpoints were identified, Metasploit modules were used to execute targeted exploitation, resulting in mixed traffic signatures of scanning bursts followed by long, high-throughput sessions. ONOS telemetry during these scenarios showed frequent topology updates and surges in packet-in messages for previously unknown hosts, along with flow-table churn consistent with the dynamic reinstallation of rules. It reflects the exploratory behaviour of adversaries mapping network surfaces before compromise and highlights the controller's visibility into reconnaissance activity through its centralised perspective.

5.3. Dataset characterisation and analytical significance

The resulting ASEADOS-SDN-IoT dataset captures a comprehensive interplay between benign IoT communication, network-layer anomalies, and SDN control reactions. Each recorded scenario encapsulates a synchronised view of three concurrent processes: (i) network traffic evolution in the data plane, (ii) ONOS decision activity in the control plane, and (iii) application-layer service responses

at IoT endpoints. This design enables fine-grained correlation between external attack manifestations and internal control feedback, a capability that traditional IDS datasets lack.

The inclusion of ONOS telemetry adds unique analytical value by exposing internal controller performance indicators. Such as flow-installation latency, packet-in burst frequency, rule churn rate, and event-queue occupancy, which together quantify the operational stress induced by attacks. This multidimensional view facilitates research into hybrid detection models that combine statistical traffic features with controller metrics, enabling detection approaches that recognise not only malicious packets but also systemic behavioural shifts in network control logic. Furthermore, the presence of both physical and virtual IoT devices introduces heterogeneity in traffic characteristics, device latency, and communication reliability. This diversity reflects the complexity of real deployments and supports the development of IDS models that generalise across device types and topologies. Each dataset segment has been rigorously labelled with time windows, attack classes, and operational states, allowing both supervised and temporal sequence learning.

By maintaining strict temporal synchronisation across data- and control-plane captures, the dataset enables novel lines of inquiry, such as correlating controller decision latency with flow congestion, or analysing the lag between attack onset and SDN response adaptation. These relationships are essential for developing resilience-aware SDN control mechanisms and for training IDS frameworks that not only detect attacks but also predict network instability before service degradation occurs.

In summary, the proposed data generation framework achieves a multi-layer representation of SDN-IoT behaviour that bridges the gap between conventional intrusion datasets and the realities of programmable networks. The combined use of Hulk, Hping3, BoNeSI, Nmap, and Metasploit ensures diversity across volumetric, coordinated, and reconnaissance attack types, while the dual benign sets establish a realistic operational baseline. By simultaneously recording IoT device traffic, host communication, and ONOS controller telemetry, the dataset captures both the external footprints of cyberattacks and their internal consequences on SDN control dynamics. This comprehensive, rigorously labelled dataset thus serves as a foundational resource for the research community-enabling the design, evaluation, and benchmarking of intelligent, cross-plane intrusion detection and mitigation mechanisms for next-generation SDN-IoT networks.

6. Dataset composition, features, and availability

The proposed ASEADOS-SDN-IoT Intrusion Dataset was developed to capture the complex interaction between SDN control mechanisms and heterogeneous IoT communication patterns under both benign and adversarial conditions. Unlike prior datasets that isolate traffic to a single domain, this dataset integrates synchronised control-plane and data-plane observations, offering a unified perspective on how IoT devices, network switches, and SDN controllers collectively respond to dynamic network events. It thereby provides an experimental foundation for studying intrusion detection, controller resilience, and adaptive mitigation strategies within programmable IoT infrastructures.

6.1. Dataset composition and structure

The dataset is organised into three synchronised subsets that collectively represent the testbed's complete operational spectrum. Each subset is captured with precise timestamp alignment, allowing temporal correlation between controller activities and IoT communications.

- **Benign-IoT Set (Data-Plane Traffic):** This subset contains all normal traffic originating from physical and virtual IoT devices, Mininet hosts, and edge servers. It captures realistic background operations, including web services (HTTP/HTTPS), file transfers (FTP), DNS lookups, remote sessions (SSH), and continuous IoT telemetry such as temperature, humidity, and motion readings. These traces represent typical low-rate IoT traffic interspersed with high-volume service communications, reflecting the heterogeneity of real-world IoT deployments.
- **Benign-ONOS Set (Control-Plane Telemetry):** It records the internal operations of the ONOS controller during stable network states. It includes packet-in, flow-mod, stats-reply, and topology-update events exchanged between ONOS and Open vSwitch (OVS). Capturing these control messages provides a behavioural baseline of controller workload and timing, enabling cross-plane correlation studies that link variations in ONOS activity with changes in network traffic.
- **Attack Set:** Four attack classes were executed, including DoS, DDoS, Botnet, and Probe-using specialised tools that generated diverse traffic behaviours. The attacks target both IoT devices and SDN elements, producing anomalies that influence flow-table states, controller latency, and link utilisation. The simultaneous collection of ONOS telemetry during these attacks provides an additional layer of insight, exposing control-plane symptoms such as packet-in bursts, rule-installation delays, and topology instability.

All packet captures were generated in **PCAP format** using tcpdump across distributed monitoring points, including the ONOS controller, IoT interfaces, and other virtual machines. The capture processes were time-synchronised to ensure microsecond-level alignment between controller actions and network responses. This synchronisation transforms the dataset from a collection of isolated traces into an integrated behavioural model of SDN-IoT operation, enabling a detailed examination of how controller decisions influence network traffic and how the network responds to those control actions in turn.

Table 6
Final labelled class distribution of the merged ASEADOS-SDN-IoT dataset.

Label	Number of Flows	Percentage (%)
Benign	260,797	57.06
DoS	127,772	27.96
DDoS	51,465	11.26
Bot	9090	1.99
Probe	7920	1.73
Total	457,044	100.00

6.2. Flow generation and labelling

To enable statistical and machine-learning analysis, the captured packet traces were converted into a structured flow-based representation using **CICFlowMeter** [94]. This conversion step organises packets that share the same communication attributes, such as source and destination IP addresses, source and destination ports, and transport-layer protocol, into unified bidirectional flows. Each flow represents a complete session between two endpoints, capturing both the direction of traffic and the session's timing behaviour. This structured representation abstracts raw packet traces into higher-level statistical summaries, enabling efficient analysis of network behaviour while significantly reducing data volume and redundancy.

CICFlowMeter extracted 83 numerical features for each flow, capturing attributes such as packet and byte counts, inter-arrival statistics, throughput, and TCP flag ratios. These features reflect multiscale behaviour—from micro-level variations in packet spacing to macro-level session characteristics. It allows for robust traffic modelling across IoT and control networks. After extraction, each flow was cross-referenced with its originating scenario and labelled as *Benign*, *DoS*, *DDoS*, *Botnet*, or *Probe*. The labelled subsets were merged into a unified CSV dataset, providing a consistent and scalable structure for training and evaluating learning-based intrusion-detection systems.

Labelling and Quality Control: Traffic capture and labelling were performed in a controlled and sequential manner. Benign traffic was collected separately under normal operating conditions, while attack traffic was captured in independent experiments, with each attack type executed individually. For every attack scenario, start and end times were explicitly defined and synchronised with packet captures and SDN controller logs to ensure accurate alignment between traffic traces and labels. Flow labels were assigned based on the capture context: all flows from benign experiments were labelled as benign, and flows from attack-specific experiments were labelled according to the corresponding attack type. Label correctness was further verified through manual inspection of randomly sampled flows using Wireshark and cross-checking with SDN controller telemetry and aggregate traffic statistics.

Table 6 summarises the distribution of flow instances in the finalised dataset, which comprises a total of 457,044 labelled records. Benign traffic accounts for the majority of the dataset, while the four attack classes collectively account for approximately 43% of the total flows. This proportional structure closely reflects real-world SDN-IoT environments, where legitimate communication dominates under regular operation, yet diverse and intermittent attacks introduce measurable anomalies. Such a balanced composition supports the development and evaluation of both traditional ML and advanced DL intrusion-detection models.

This flow representation reduces data volume while preserving discriminative statistical signatures, ensures compatibility with standard ML/DL frameworks, and supports time-series reconstruction for temporal analysis. The resulting dataset thus combines interpretability with analytical depth, enabling cross-layer correlation, behavioural profiling, and predictive modelling.

6.3. Feature design rationale

The proposed dataset focuses on statistical flow-level features extracted from SDN-IoT traffic, selected based on practical deployment and methodological considerations. Statistical features can be computed in real time at the SDN controller or monitoring plane without requiring long observation windows or stateful per-device tracking, which aligns well with SDN-IoT environments where timely detection and controller-level visibility are critical. Moreover, statistical features are largely controller-agnostic and protocol-independent, enabling reproducibility across different SDN controllers, IoT protocols, and deployment scenarios.

In contrast, temporal features often depend on fixed observation windows or sequential dependencies, while domain-specific features may rely on device semantics or application-layer knowledge that is not consistently available across heterogeneous IoT deployments. Such dependencies can limit dataset generalisability and hinder fair benchmarking. The use of statistical features also supports compatibility with resource-constrained edge and IoT environments, including lightweight and TinyML-based intrusion detection models. Temporal, spectral, or cross-domain features, therefore, represent promising directions for future dataset extensions rather than omissions in the current design.

6.4. Extracted features

Table 7 summarises the 83 extracted flow-level features, which quantify the structural and temporal behaviour of bidirectional flows, including packet counts, timing irregularities, and protocol-state variations. The features are grouped into categories, includ-

Table 7
Extracted flow-level features and descriptions.

#	Feature	Description	#	Feature	Description
1	Flow ID	Unique flow identifier.	43	Fwd Pkts/s	Forward packet rate.
2	Src IP	Source IP address.	44	Bwd Pkts/s	Backward packet rate.
3	Src Port	Source transport port.	45	Pkt Len Min	Minimum packet length.
4	Dst IP	Destination IP address.	46	Pkt Len Max	Maximum packet length.
5	Dst Port	Destination transport port.	47	Pkt Len Mean	Mean packet length.
6	Protocol	Transport protocol number.	48	Pkt Len Std	Std.dev. of packet length.
7	Timestamp	Flow start time.	49	Pkt Len Var	Variance of packet length.
8	Flow Duration	Duration of the flow.	50	FIN Flag Cnt	Count of FIN flags.
9	Tot Fwd Pkts	Total forward packets.	51	SYN Flag Cnt	Count of SYN flags.
10	Tot Bwd Pkts	Total backward packets.	52	RST Flag Cnt	Count of RST flags.
11	TotLen Fwd Pkts	Total forward bytes.	53	PSH Flag Cnt	Count of PSH flags.
12	TotLen Bwd Pkts	Total backward bytes.	54	ACK Flag Cnt	Count of ACK flags.
13	Fwd Pkt Len Max	Max forward packet length.	55	URG Flag Cnt	Count of URG flags.
14	Fwd Pkt Len Min	Min forward packet length.	56	CWE Flag Cnt	Count of CWR flags.
15	Fwd Pkt Len Mean	Mean forward packet length.	57	ECE Flag Cnt	Count of ECE flags.
16	Fwd Pkt Len Std	Std. dev. f/ward packet length.	58	Down/Up Ratio	Download to upload ratio.
17	Bwd Pkt Len Max	Max backward packet length.	59	Pkt Size Avg	Average packet size.
18	Bwd Pkt Len Min	Min backward packet length.	60	Fwd Seg Size Avg	Avg. forward segment size.
19	Bwd Pkt Len Mean	Mean backward packet length.	61	Bwd Seg Size Avg	Avg. backward segment size.
20	Bwd Pkt Len Std	Std. dev. b/ward packet length.	62	Fwd Byts/b Avg	Bytes per forward bulk.
21	Flow Byts/s	Average bytes per second.	63	Fwd Pkts/b Avg	Packets per forward bulk.
22	Flow Pkts/s	Average packets per second.	64	Fwd Blk Rate Avg	Rate of forward bulk.
23	Flow IAT Mean	Mean inter-arrival time.	65	Bwd Byts/b Avg	Bytes per backward bulk.
24	Flow IAT Std	Std. dev. of inter-arrival time.	66	Bwd Pkts/b Avg	Packets per backward bulk.
25	Flow IAT Max	Max inter-arrival time.	67	Bwd Blk Rate Avg	Rate of backward bulk.
26	Flow IAT Min	Min inter-arrival time.	68	Subflow Fwd Pkts	Packets in forward subflows.
27	Fwd IAT Tot	Total forward IAT.	69	Subflow Fwd Byts	Bytes in forward subflows.
28	Fwd IAT Mean	Mean forward IAT.	70	Subflow Bwd Pkts	Packets in backward subflows.
29	Fwd IAT Std	Std. dev. of forward IAT.	71	Subflow Bwd Byts	Bytes in backward subflows.
30	Fwd IAT Max	Max forward IAT.	72	Init Fwd Win Byts	Initial forward window size.
31	Fwd IAT Min	Min forward IAT.	73	Init Bwd Win Byts	Initial backward window size.
32	Bwd IAT Tot	Total backward IAT.	74	Fwd Act Data Pkts	Forward packets with ≥ 1 byte.
33	Bwd IAT Mean	Mean backward IAT.	75	Fwd Seg Size Min	Min forward segment size.
34	Bwd IAT Std	Std. dev. of backward IAT.	76	Active Mean	Mean active time.
35	Bwd IAT Max	Max backward IAT.	77	Active Std	Std. dev. of active time.
36	Bwd IAT Min	Min backward IAT.	78	Active Max	Max active time.
37	Fwd PSH Flags	Count of forward PSH flags.	79	Active Min	Min active time.
38	Bwd PSH Flags	Count of backward PSH flags.	80	Idle Mean	Mean idle time.
39	Fwd URG Flags	Count of forward URG flags.	81	Idle Std	Std. dev. of idle time.
40	Bwd URG Flags	Count of backward URG flags.	82	Idle Max	Max idle time.
41	Fwd Header Len	Forward header length.	83	Idle Min	Min idle time.
42	Bwd Header Len	Backward header length.	84	Label	Traffic label.

ing flow statistics, inter-arrival times, packet-length distributions, and TCP flag counters, which enable fine-grained statistical and machine-learning analysis.

The selected feature set captures multi-layered characteristics of SDN-IoT communication. Inter-arrival metrics and active/idle time distributions reveal the temporal rhythm of IoT sensor transmissions, while byte- and packet-oriented measures expose throughput asymmetry typical of upstream telemetry. TCP flag counts and window-size indicators reflect connection management behaviour and flow-rule adjustments triggered by the SDN controller. By combining these temporal, statistical, and protocol-level perspectives, the dataset enables high-resolution modelling of traffic behaviour. It supports both traditional ML classifiers and sequence-based DL architectures. Each flow instance retains interpretable semantics while remaining compatible with large-scale analytics, providing an empirical foundation for the experimental evaluation in the next section.

6.5. Dataset availability and research utility

The finalised dataset is publicly available and structured to facilitate reproducible research. It consists of three core components:

- **Raw PCAP archives** (approx. 35 GB uncompressed, approx. 13 GB compressed): Complete packet-level captures for all benign and attack experiments, recorded using `tcpdump` at distributed vantage points, including the ONOS controller, IoT interfaces, and other virtual machines. This multi-point capture strategy ensures temporal alignment between control-plane events and data-plane packets. Time synchronisation across nodes enables accurate reconstruction of cause-and-effect relationships between controller actions and traffic anomalies.

Table 8
Selected cleaned flow-level features used in the final dataset.

#	Feature	#	Feature	#	Feature
1	Protocol	23	Fwd IAT Std	45	ACK Flag Cnt
2	Flow Duration	24	Fwd IAT Max	46	Down/Up Ratio
3	Tot Fwd Pkts	25	Fwd IAT Min	47	Pkt Size Avg
4	Tot Bwd Pkts	26	Bwd IAT Tot	48	Fwd Seg Size Avg
5	TotLen Fwd Pkts	27	Bwd IAT Mean	49	Bwd Seg Size Avg
6	TotLen Bwd Pkts	28	Bwd IAT Std	50	Subflow Fwd Pkts
7	Fwd Pkt Len Max	29	Bwd IAT Max	51	Subflow Fwd Byts
8	Fwd Pkt Len Min	30	Bwd IAT Min	52	Subflow Bwd Pkts
9	Fwd Pkt Len Mean	31	Bwd PSH Flags	53	Subflow Bwd Byts
10	Fwd Pkt Len Std	32	Fwd Header Len	54	Init Bwd Win Byts
11	Bwd Pkt Len Max	33	Bwd Header Len	55	Fwd Act Data Pkts
12	Bwd Pkt Len Min	34	Fwd Pkts/s	56	Active Mean
13	Bwd Pkt Len Mean	35	Bwd Pkts/s	57	Active Std
14	Bwd Pkt Len Std	36	Pkt Len Min	58	Active Max
15	Flow Byts/s	37	Pkt Len Max	59	Active Min
16	Flow Pkts/s	38	Pkt Len Mean	60	Idle Mean
17	Flow IAT Mean	39	Pkt Len Std	61	Idle Std
18	Flow IAT Std	40	Pkt Len Var	62	Idle Max
19	Flow IAT Max	41	FIN Flag Cnt	63	Idle Min
20	Flow IAT Min	42	SYN Flag Cnt	64	Label
21	Fwd IAT Tot	43	RST Flag Cnt		
22	Fwd IAT Mean	44	PSH Flag Cnt		

- **Processed CSV dataset:** Generated using **CICFlowMeter**, this file includes 83 extracted features for every bidirectional flow, labelled according to the originating scenario (*Benign*, *DoS*, *DDoS*, *Botnet*, *Probe*). The unified structure enables both binary and multi-class classification, as well as research on temporal correlation and anomaly detection.
- **Documentation and configuration files:** Supplementary resources, including dataset description and other helpful information.

All captured traces were validated in **Wireshark**, confirming timestamp integrity and flow continuity. The dataset's integration of physical IoT communication, virtual network dynamics, and SDN control telemetry creates a realistic environment for advanced research. It enables a wide range of investigations, including:

- Flow-based intrusion detection and hybrid ML model evaluation,
- Cross-plane correlation between ONOS decision processes and IoT traffic responses,
- Performance and scalability analysis of SDN controllers under stress,
- Temporal modelling of IoT communication behaviour, and
- Adaptive policy design for proactive threat mitigation in programmable networks.

By aligning flow-level semantics with control-plane context, this dataset establishes a novel research foundation for analysing cyber-resilience and intelligent orchestration in SDN-IoT systems.

Sections 5 and 6 presented the design, construction, and analytical structure of the proposed **ASEADOS-SDN-IoT Intrusion Dataset**. Through its synchronised capture design, 83-feature flow schema, and public accessibility, the dataset offers a comprehensive view of programmable network behaviour under both benign and adversarial conditions. It serves as a robust empirical platform for developing and benchmarking intelligent intrusion-detection and mitigation strategies. The following section presents the **experimental evaluation**, demonstrating the dataset's representativeness and analytical potential through quantitative assessment.

7. Experimental evaluation

This section presents the experimental evaluation conducted to validate the analytical quality and practical applicability of the proposed ASEADOS-SDN-IoT Intrusion Dataset. All experiments were performed in a high-performance Linux environment on a dedicated server equipped with dual Intel Xeon Gold 6140 CPUs (72 cores) and 256 GB of system memory, running Ubuntu 20.04. This computational infrastructure provided robust parallel processing for feature extraction, preprocessing, and large-scale learning, without introducing performance bottlenecks. The evaluation focuses on examining data consistency, statistical behaviour, and the dataset's suitability for training intelligent intrusion-detection models. All experimental procedures were implemented in Python 3.10 using the NumPy, pandas, and scikit-learn libraries within a Jupyter environment.

7.1. Dataset overview

The dataset integrates traffic from both the SDN control plane and the IoT data plane, covering a wide range of benign and malicious activities. It consists of five labelled traffic classes. Including *Benign*, *DoS*, *DDoS*, *Botnet*, and *Probe* that captured under

Table 9
Comparison of dataset composition before and after cleaning.

Label	Before Cleaning (Flows)	Before (%)	After Cleaning (Flows)	After (%)
Benign	260,797	57.06	233,745	54.74
DoS	127,772	27.96	126,506	29.62
DDoS	51,465	11.26	50,269	11.77
Bot	9090	1.99	8957	2.10
Probe	7920	1.73	7564	1.77
Total	457,044	100.00	427,041	100.00

diverse network states. After merging and validation, the dataset contained 457,044 flow records, which were reduced to 427,041 high-quality flows following a standard cleaning process. This refined dataset maintains the temporal contact between control- and data-plane events, enabling precise behavioural correlation across layers.

Table 9 summarises the number of flows before and after cleaning. Benign traffic remains the majority class, representing typical IoT and ONOS control communication, while attack flows collectively account for approximately 45% of total records. This distribution provides a realistic traffic mix, reflecting practical SDN-IoT networks where normal operations dominate but are periodically disrupted by adversarial events.

7.2. Data preprocessing

The preprocessing pipeline was carefully designed to ensure the dataset's integrity, minimise noise, and prepare it for downstream machine-learning applications. The steps are summarised below:

- **Data Cleaning:** The raw flow files filtered to remove duplicate entries, incomplete rows, and flows with zero duration or missing values. This step eliminated corrupted or inactive sessions that could bias feature distributions or reduce classifier generalisation.
- **Feature Refinement:** From the 83 originally extracted attributes, 63 were retained after discarding identifier fields and features exhibiting near-zero variance or multicollinearity. The retained attributes capture statistical, temporal, and protocol-level characteristics of bidirectional flows, enabling both behavioural and volumetric analysis.
- **Outlier and Consistency Control:** Flows with abnormally high durations, packet counts, or throughput values were clipped to prevent extreme-value distortion. It ensured that traffic variability reflected realistic IoT and SDN conditions rather than capturing noise or measurement anomalies.
- **Normalization:** All numeric features were scaled using Min–Max normalization to a common [0,1] range. This transformation adjusts feature importance, allowing fair comparison across metrics such as packet counts, inter-arrival times, and byte volumes.
- **Label Encoding:** The categorical labels were mapped to numerical integers as follows: 0 (*Benign*), 1 (*DoS*), 2 (*DDoS*), 3 (*Botnet*), and 4 (*Probe*). This encoding enabled seamless integration with supervised learning algorithms.
- **Data Partitioning:** The processed dataset was randomly divided using an 80–20 stratified split, preserving proportional representation across all five classes. The resulting subsets contain 341,632 flows for training and 85,409 for testing, maintaining statistical balance between benign and attack traffic.

The preprocessing pipeline produced a clean, numerically stable dataset optimised for statistical characterisation and predictive modelling. This carefully innovated approach enables a fair comparison of learning algorithms and behaviour analysis, as explored in the subsequent evaluation stages.

7.3. Statistical validation and dataset novelty (preprocessed)

To further substantiate dataset novelty, statistical validation was performed on the extracted feature-level data. One-way analysis of variance (ANOVA) with false discovery rate (FDR) correction confirms that all extracted features exhibit statistically significant inter-class differences ($p < .05$), with several features demonstrating large effect sizes (e.g., protocol and TCP flag features with $\eta^2 > 0.3$), indicating strong discriminative capability. In addition, multivariate analysis using principal component analysis (PCA) reveals clear structural differentiation between benign and malicious traffic, as well as among different attack categories, with the first two principal components explaining approximately 33% of the total variance.

Together, the design rationale and statistical evidence demonstrate that the proposed dataset captures meaningful and diverse SDN-IoT traffic characteristics. By combining real-time-extractable statistical features with quantitative validation of feature discriminativeness, the dataset provides a robust and novel benchmark for training and evaluating intrusion detection systems in SDN-IoT environments. Table 10 summarises the key design choices and statistical validation results.

7.4. Analysis techniques and metrics

This study employs both ML and DL approaches to analyse network behaviour within the SDN-IoT environment. The dual framework enables comparison between traditional statistical models and representation-based neural architectures, providing complementary insights into intrusion characteristics and model adaptability. We defined five binary classification scenarios to reflect realistic

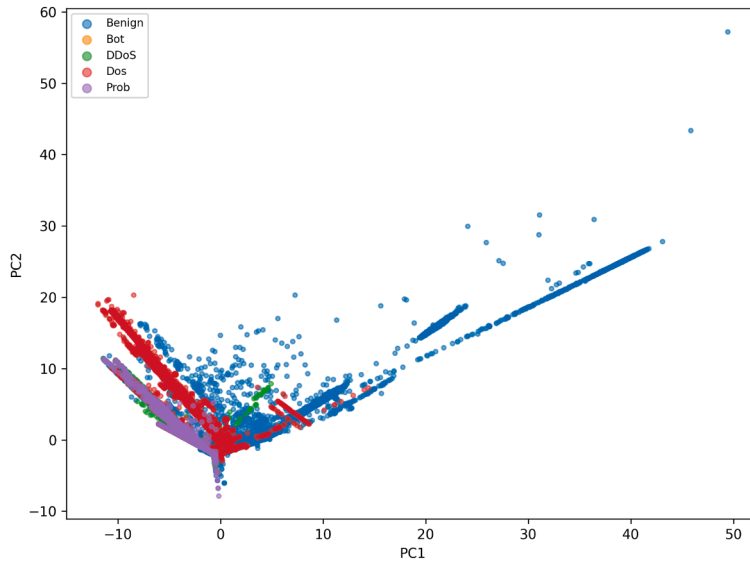


Fig. 8. PCA-based visualisation of the joint feature space for benign and malicious traffic in the ASEADOS-SDN-IoT dataset.

Table 10

Summary of dataset design rationale and statistical validation.

Aspect	Evidence and Justification
Feature domain	Statistical flow-level features selected for real-time extractability, controller-agnostic deployment, and compatibility with resource-constrained SDN-IoT environments.
Feature relevance	One-way ANOVA with FDR correction shows 63/63 features exhibit statistically significant inter-class differences ($p < .05$).
Effect size	Several features demonstrate large effect sizes (e.g., protocol and TCP flag features with $\eta^2 > 0.3$), indicating strong discriminative power.
Multivariate structure	PCA analysis shows clear separation between benign and attack traffic in the joint feature space, with the first two components explaining approximately 33% of variance.
Dataset novelty	Cross-layer SDN-IoT traffic characteristics captured through statistically validated features, enabling realistic benchmarking of IDS models.

detection conditions. Benign traffic was contrasted separately with DoS, DDoS, Botnet, and Probe attacks, while a fifth scenario combined all attacks to represent a generalised intrusion setting. This design supports consistent evaluation across focused and mixed threat environments, enabling a comprehensive assessment of each model's robustness and detection capability.

7.4.1. ML models

The ML analysis encompassed a diverse spectrum of algorithms that reflect different decision paradigms. Decision Tree and Random Forest capture hierarchical relationships through recursive partitioning; K-Nearest Neighbours relies on local instance proximity; Gaussian Naïve Bayes models statistical independence between attributes; Support Vector Machine establishes optimal separating hyperplanes; and ensemble methods such as AdaBoost, LightGBM, and CatBoost enhance classification stability through iterative boosting and feature-weighted voting.

In this study, we selected these algorithms to examine how traditional learning mechanisms respond to heterogeneous IoT traffic distributions. Their interpretability, computational efficiency, and relative maturity in network analysis make them suitable baselines for evaluating the proposed dataset's structural soundness and its capacity to generalise across multiple intrusion patterns. The comparative evaluation of these algorithms also facilitates understanding of the trade-offs between model complexity, decision transparency, and detection scalability.

7.4.2. DL Models

To complement classical methods, DL architectures were employed to capture complex, high-level relationships among features in the dataset. The Multi-Layer Perceptron (MLP) captures global non-linear interactions between features. At the same time, recurrent

Table 11

Summary of classical ML models and configurations used for experiments.

Model	Type	Key Settings
Decision Tree (DT)	Tree-based classifier	Gini criterion, depth tuned via CV
Random Forest (RF)	Ensemble (bagging)	100 estimators, max features = sqrt
K-Nearest Neighbours (KNN)	Instance-based learner	$k = 5$, Euclidean distance
Gaussian Naive Bayes (GNB)	Probabilistic model	Gaussian assumption per feature
AdaBoost	Ensemble (boosting)	100 estimators, learning rate = 0.1
Support Vector Machine (SVM)	Margin-based classifier	RBF kernel, C and γ tuned via CV
LightGBM	Gradient boosting trees	100 estimators, learning rate = 0.05
CatBoost	Ordered boosting trees	Depth = 6, learning rate = 0.05

Table 12

Summary of DL architectures and configurations used for experiments.

Model	Main Layers	Key Settings
MLP	Dense(128, ReLU) → Dropout(0.3) → Dense(64, ReLU) → Dropout(0.3) → Dense(1, Sigmoid)	Adam optimizer; binary cross-entropy loss; 20 epochs; batch size 128
LSTM	LSTM(64) → Dropout(0.3) → Dense(1, Sigmoid)	Adam optimizer; binary cross-entropy loss; 20 epochs; batch size 128
GRU	GRU(64) → Dropout(0.3) → Dense(1, Sigmoid)	Adam optimizer; binary cross-entropy loss; 20 epochs; batch size 128
CNN	Conv1D(32, kernel=3, ReLU) → Max-Pooling1D(2) → Flatten → Dropout(0.3) → Dense(64, ReLU) → Dropout(0.3) → Dense(1, Sigmoid)	Adam optimizer; binary cross-entropy loss; 20 epochs; batch size 128
CNN + LSTM	Conv1D(32, kernel=3, ReLU) → Max-Pooling1D(2) → LSTM(64) → Dropout(0.3) → Dense(1, Sigmoid)	Adam optimizer; binary cross-entropy loss; 20 epochs; batch size 128

models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are designed to extract temporal dependencies indicative of evolving attack dynamics. The Convolutional Neural Network (CNN) focuses on local spatial patterns among correlated statistical attributes, and the hybrid CNN-LSTM integrates spatial abstraction with temporal modelling to achieve a holistic representation of intrusions.

These architectures enable the analysis of behavioural continuity in network flows, revealing whether deep temporal abstraction enhances discriminative performance relative to static learning. Their inclusion also examines scalability and adaptability under dynamic traffic, reflecting their relevance to deployment in complex SDN-IoT infrastructures.

7.4.3. Evaluation metrics

Model assessment was conducted through a unified set of quantitative indicators that capture multiple aspects of detection performance. This study used **Accuracy**, **precision**, **recall**, **F1-score**, and **false-positive rate (FPR)** for both ML and DL models to measure correctness, sensitivity, and classification reliability. In addition, the area under the receiver operating characteristic curve (ROC-AUC) was used exclusively for DL models to evaluate probabilistic discrimination across varying decision thresholds.

Together, these metrics provide a balanced analytical perspective encompassing both operational reliability and theoretical discriminative power. Accuracy reflects global classification correctness; precision and recall highlight the model's ability to control false alarms and missed detections; the F1-score represents their harmonic balance; FPR evaluates the rate of benign misclassification—a critical aspect in SDN-IoT control environments; and ROC-AUC quantifies threshold-independent separability for probabilistic models. This metric configuration establishes a multidimensional assessment framework that captures not only detection accuracy but also robustness and practicality for deployment.

7.5. Results and discussion

The experimental results provide a comprehensive understanding of how the proposed ASEADOS-SDN-IoT dataset supports intelligent intrusion detection through both classical ML and DL paradigms. The diversity of attack categories, multi-layered traffic representation, and balanced feature composition enable accurate classification across a wide range of network behaviours. The results demonstrate not only the dataset's discriminative potential but also its capacity to reveal the strengths and limitations of different analytical techniques under varying traffic dynamics.

7.5.1. ML models

The ML analysis establishes a strong foundational benchmark for evaluating the effectiveness of traditional supervised algorithms on the proposed dataset. The results across all scenarios show that the dataset maintains clear separability between benign and

Table 13

Performance metrics of the ML models on the ASEADOS-SDN-IoT dataset.

Attack	Key metrics	ML model							
		DT	RF	KNN	Gaus. NB	AdaBoost	SVM	LightGBM	CatBoost
DoS	Accuracy	0.9455	0.9467	0.9444	0.9112	0.9550	0.9258	0.9555	0.9550
	Precision	0.9444	0.947	0.9425	0.9216	0.9621	0.9326	0.9639	0.9623
	Recall	0.9467	0.9463	0.9471	0.9112	0.9473	0.9180	0.9464	0.9472
	F1-score	0.9455	0.9467	0.9448	0.9106	0.9546	0.9252	0.9551	0.9547
	FPR	0.0551	0.0537	0.0584	0.0103	0.0372	0.0662	0.0352	0.0370
DDoS	Accuracy	0.8975	0.9014	0.9132	0.9167	0.9328	0.9312	0.9349	0.9326
	Precision	0.8981	0.8980	0.8862	0.9246	0.8893	0.8849	0.8927	0.8891
	Recall	0.8968	0.9056	0.9496	0.9167	0.9887	0.9913	0.9885	0.9885
	F1-score	0.8974	0.9018	0.9168	0.9163	0.9364	0.9351	0.9382	0.9362
	FPR	0.1017	0.1030	0.1237	0.1518	0.1230	0.1290	0.1187	0.1230
Bot	Accuracy	0.9595	0.9606	0.9676	0.9294	0.9704	0.9646	0.9729	0.9726
	Precision	0.9577	0.9573	0.9525	0.9358	0.9469	0.9669	0.9539	0.9519
	Recall	0.9615	0.9643	0.9844	0.9294	0.9966	0.9646	0.9939	0.9955
	F1-score	0.9596	0.9608	0.9681	0.9291	0.9712	0.9645	0.9735	0.9733
	FPR	0.0423	0.0436	0.0497	0.1311	0.0551	0.0709	0.0488	0.0500
Prob	Accuracy	0.9207	0.9276	0.9293	0.9157	0.9392	0.9243	0.9415	0.9415
	Precision	0.9207	0.9212	0.9166	0.9279	0.9099	0.8826	0.9185	0.9129
	Recall	0.9207	0.9352	0.9445	0.9157	0.9749	0.9788	0.9689	0.9762
	F1-score	0.9207	0.9282	0.9303	0.9151	0.9413	0.9282	0.9431	0.9435
	FPR	0.0793	0.0800	0.0869	0.1685	0.0965	0.1301	0.086	0.0931
All attacks	Accuracy	0.9108	0.9138	0.9163	0.8851	0.9323	0.9171	0.9328	0.9325
	Precision	0.9101	0.9130	0.8973	0.8889	0.9316	0.9177	0.9328	0.9321
	Recall	0.9099	0.9130	0.9206	0.8851	0.9354	0.9214	0.9368	0.9360
	F1-score	0.9100	0.9130	0.9088	0.8854	0.9321	0.9170	0.9327	0.9323
	FPR	0.0802	0.0780	0.0876	0.1441	0.0972	0.1240	0.1044	0.1006

malicious flows while preserving a realistic degree of overlap. This balance challenges model generalisation and represents a crucial characteristic of a practical intrusion detection benchmark.

For the **DoS detection scenario**, ensemble-based models such as AdaBoost, LightGBM, and CatBoost achieved the highest overall accuracy and F1-scores (≈ 0.95), confirming the dataset's ability to represent flooding behaviour through discriminative flow-level statistics (See Fig. 9). These models exploit hierarchical and boosting-based feature weighting to effectively isolate volumetric anomalies generated during DoS events. The consistent performance of ensemble learners also indicates that the dataset provides clear statistical signatures of high-rate packet generation and protocol imbalance. The **DDoS scenario** demonstrated slightly lower accuracy across all classifiers (≈ 0.90 - 0.93), yet the recall values remained remarkably high for ensemble and kernel-based methods. It highlights the dataset's realism in capturing distributed attack behaviour, where multiple concurrent flows blur conventional decision boundaries. The results show that models such as SVMs and boosting ensembles successfully generalise across distributed patterns, but also exhibit marginally higher false positives due to the similarity between multi-source attack flows and legitimate high-traffic events in SDN-IoT environments.

In the **Botnet scenario**, the dataset's temporal and structural fidelity allowed all models to achieve their strongest results, with LightGBM and CatBoost surpassing 0.97 in accuracy and F1-score. The high recall observed here validates the dataset's inclusion of low-rate, coordinated bot activity-traffic that often escapes detection in traditional IoT datasets. It confirms that the dataset captures command-and-control propagation patterns with sufficient diversity for robust model training. For **Probe traffic**, most models sustained high detection rates (≈ 0.92 - 0.94 accuracy). The balanced performance across tree-based and distance-based methods demonstrates that the dataset maintains distinguishable reconnaissance patterns without oversimplifying feature correlations. The feature composition enables both local and global statistical models to identify scanning and exploration traffic with precision, reinforcing the dataset's representational completeness.

The **combined all-attack scenario** presents the most complex evaluation due to mixed traffic behaviours and overlapping features. Despite this, LightGBM and AdaBoost maintained superior accuracy (≈ 0.93) and balanced precision-recall scores, reflecting their resilience to feature heterogeneity. The slight rise in false positives (≈ 0.10) shows the challenge of generalising across diverse intrusions, a condition deliberately maintained to ensure realistic model assessment (See Fig. 10).

Overall, the ML evaluation confirms that the proposed dataset provides a meaningful balance between discriminative separability and structural realism. Ensemble algorithms excel because they can exploit nonlinear feature relations derived from SDN control-plane interactions and IoT application traffic. Simpler algorithms, though more interpretable, struggled with the dataset's overlapping feature regions. It indicates that the dataset accurately reflects operational noise and the natural variability of legitimate network traffic.

7.5.2. DL models

In the next step, DL models are explored to capture complex nonlinear relationships and temporal dependencies that traditional algorithms may overlook. The results demonstrate that the proposed dataset effectively supports these architectures, delivering clear improvements in accuracy, recall, and F1-score across all attack scenarios. The diverse features, particularly those capturing temporal

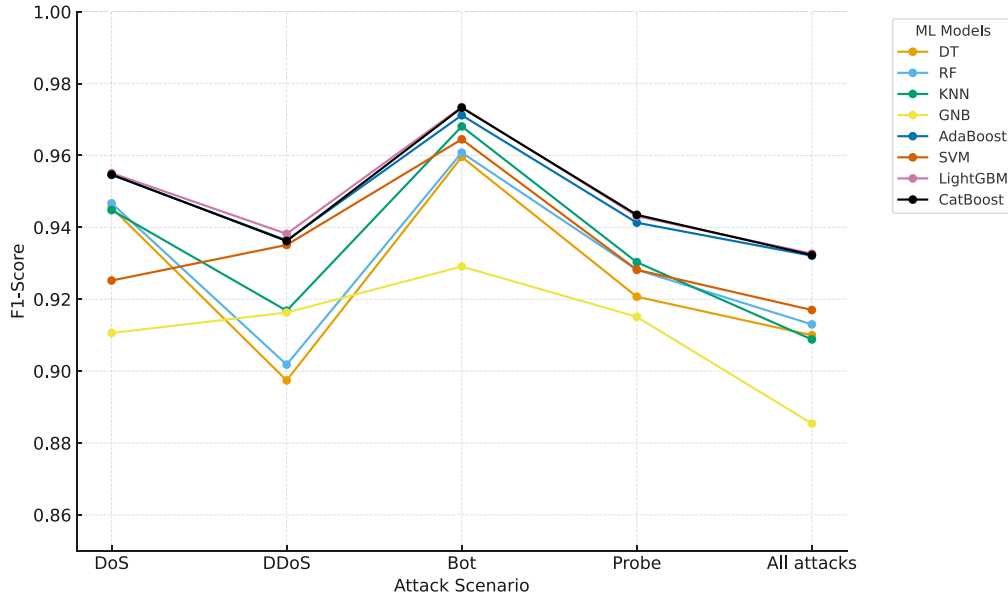


Fig. 9. Comparison of ML models by F1-score across attack scenarios.

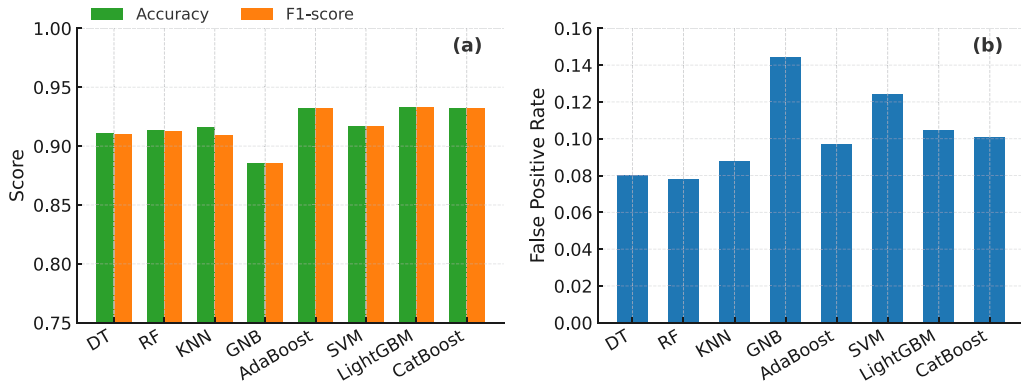


Fig. 10. Comparative performance of ML models on the All-Attack scenario of the proposed ASEADOS-SDN-IoT dataset. Panel (a) illustrates accuracy and F1-scores across models, while panel (b) presents corresponding false-positive rates.

flow dynamics and controller interactions, enable neural models to identify subtle intrusion patterns that static classifiers often miss. It confirms that the dataset's design successfully bridges the gap between data-plane and control-plane information, allowing deeper representations of SDN-IoT traffic patterns.

The **MLP** baseline achieved stable, competitive results across all scenarios, indicating that even shallow neural networks can leverage the dataset's structured, informative feature representation. However, sequential architectures such as **LSTM** and **GRU** delivered notable improvements, especially for the DDoS and all-attack scenarios, where recall exceeded 0.98. These models benefited from the temporal coherence embedded in the dataset, where packet timing, duration, and inter-arrival statistics provide meaningful context for identifying evolving attack sequences. The **CNN** architecture performed exceptionally well in detecting DoS and Probe activities by learning spatial correlations among aggregated flow features. Its high precision and F1-scores (>0.95) indicate that the proposed dataset preserves strong spatial continuity between related statistical attributes, allowing effective convolutional representation. Building on this, the **hybrid CNN-LSTM** framework delivered the best overall performance, achieving F1-scores near 0.98 and ROC-AUC values above 0.99 across most attack types. This improvement shows that the dataset provides sufficient temporal depth and spatial granularity to support integrated spatio-temporal learning. By combining flow-level statistics with sequential dependencies, the hybrid network detects both short-term anomalies and long-duration attack behaviours. Such comprehensive detection capability is rarely achieved in existing IoT intrusion datasets.

Another notable observation is the consistent reduction in false-positive rates (≤ 0.06) across all deep architectures. This outcome indicates that the dataset maintains well-defined boundaries between benign and attack traffic, even after nonlinear transformations

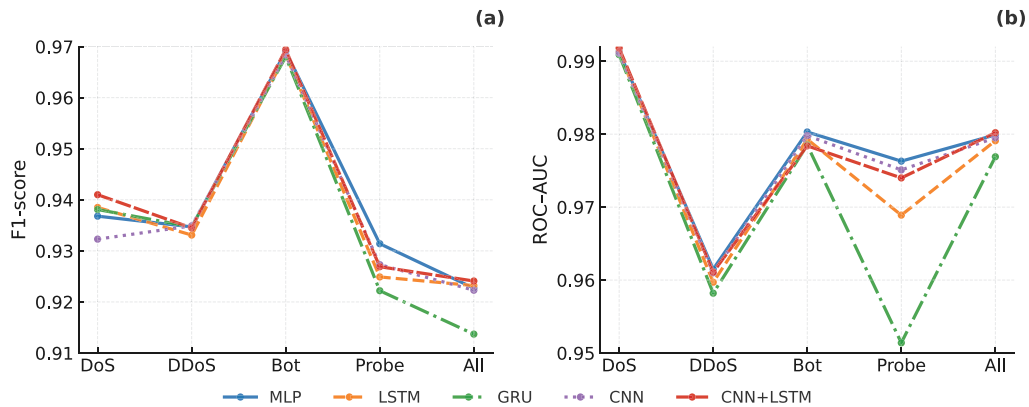


Fig. 11. Comparison of DL models by F1-score and ROC-AUC across attack scenarios. (a) F1-score trends; (b) ROC-AUC performance.

Table 14

Performance metrics of the DL models on the ASEADOS-SDN-IoT dataset.

Attack	Key metrics	DL model MLP	LSTM	GRU	CNN	Hybrid CNN + LSTM
DoS	Accuracy	0.9358	0.9383	0.9386	0.9331	0.9413
	Precision	0.9220	0.9354	0.9457	0.9434	0.9462
	Recall	0.9521	0.9471	0.9306	0.9214	0.9353
	F1-score	0.9368	0.9385	0.9381	0.9323	0.9410
	FPR	0.0803	0.0652	0.0539	0.0553	0.0532
DDoS	ROC-AUC	0.9913	0.9912	0.9909	0.9911	0.9918
	Accuracy	0.9308	0.9296	0.9307	0.9309	0.9308
	Precision	0.8846	0.8885	0.8839	0.8832	0.8877
	Recall	0.9908	0.9825	0.9917	0.9931	0.9864
	F1-score	0.9347	0.9331	0.9347	0.9349	0.9345
Bot	FPR	0.1297	0.1235	0.1308	0.1310	0.1246
	ROC-AUC	0.9615	0.9597	0.9582	0.9610	0.9611
	Accuracy	0.9682	0.9673	0.9668	0.9673	0.9685
	Precision	0.9406	0.9387	0.9382	0.9387	0.9407
	Recall	0.9994	1.0000	0.9994	1.0000	1.0000
Prob	F1-score	0.9691	0.9684	0.9678	0.9684	0.9694
	FPR	0.0630	0.0651	0.0660	0.0651	0.0630
	ROC-AUC	0.9803	0.9793	0.9785	0.9798	0.9784
	Accuracy	0.9283	0.9223	0.9164	0.923	0.9213
	Precision	0.8923	0.8954	0.8637	0.8787	0.8653
All attacks	Recall	0.9742	0.9564	0.9888	0.9815	0.9980
	F1-score	0.9314	0.9249	0.922	0.9273	0.9269
	FPR	0.1171	0.1115	0.1555	0.1352	0.1551
	ROC-AUC	0.9763	0.9689	0.9514	0.9751	0.9740
	Accuracy	0.9228	0.9234	0.9140	0.9225	0.9242
	Precision	0.9238	0.9241	0.9132	0.9222	0.9249
	Recall	0.9275	0.9278	0.9168	0.9260	0.9287
	F1-score	0.9227	0.9232	0.9137	0.9223	0.9241
	FPR	0.1220	0.1194	0.1124	0.1115	0.1182
	ROC-AUC	0.9799	0.9791	0.9769	0.9795	0.9802

in high-dimensional feature spaces. The inclusion of ROC-AUC as an additional metric further validates the strong separability of classes, with most models approaching near-perfect discrimination (See Figs. 11 and 12).

Overall, the DL results highlight the dataset's suitability as a modern benchmark for intelligent SDN-IoT intrusion detection. By integrating realistic temporal and spatial characteristics from both the control and data planes, the dataset enables deep architectures to learn transferable, generalizable intrusion representations. Its balanced diversity across flooding, reconnaissance, and botnet behaviours provides a more faithful reflection of real-world network dynamics than existing IoT intrusion datasets, which often lack temporal granularity or rely on synthetic traffic patterns.

7.5.3. Comparative insights

The comparative analysis between ML and DL models highlights the strengths of the ASEADOS-SDN-IoT dataset. Classical ML algorithms such as Random Forest and SVM achieved interpretable and consistent accuracy across all attack categories, confirming the dataset's balanced distribution, high feature separability, and structural quality. This stability demonstrates that ASEADOS-SDN-

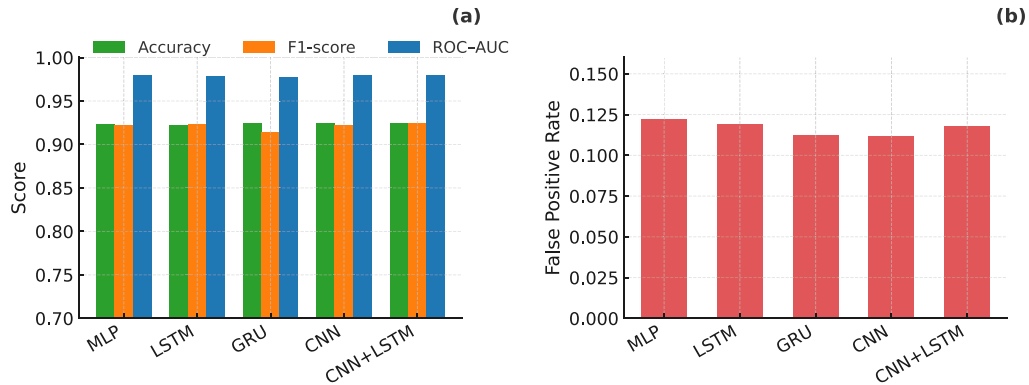


Fig. 12. Deep learning model performance on the All-Attack scenario. (a) Accuracy, F1-score, and ROC-AUC. (b) False-positive rate.

IoT preserves statistical realism and minimises the redundancy and bias common in synthetic or replay-based datasets used in prior SDN-IoT research.

In contrast, DL architectures benefited significantly from ASEADOS-SDN-IoT's temporal continuity, packet-flow diversity, and cross-layer composition. CNN, LSTM, and DNN models effectively captured spatial-temporal dependencies in IoT traffic flows and SDN control-plane signals, learning richer feature hierarchies than those from purely data-plane datasets. The enhanced recall, precision, and ROC-AUC achieved by these models validate the dataset's ability to reveal latent behavioural dependencies, such as controller adaptation to traffic surges, flow reinstallation frequency, and packet-in response timing, which remain invisible in synthetic datasets. This deeper correlation between control reactions and packet-flow dynamics enables more accurate modelling of network state transitions and the propagation of intrusions across the SDN-IoT feedback loop.

A defining strength of ASEADOS-SDN-IoT is its authentic representation of real-time SDN-IoT feedback mechanisms. Whereas earlier datasets rely on artificially generated traffic or post-processed controller logs, ASEADOS-SDN-IoT synchronously captures actual packet flows, flow-rule installations, and packet-in events as they occur within a live ONOS-controlled environment. This fidelity enables DL models to learn both the characteristics of malicious traffic and the semantics of responses-how the controller's decision logic evolves as threats emerge. Such an integrated context enhances temporal reasoning in LSTM architectures and spatial abstraction in CNNs, leading to more stable gradients and reduced loss fluctuations during training. Empirical evidence further supports the dataset's optimisation advantages. Models trained on ASEADOS-SDN-IoT exhibit smoother and faster convergence of the cross-entropy loss, lower epoch-to-epoch variance, and improved generalisation. It addresses the unstable optimisation behaviour observed in synthetic SDN-IoT datasets, where the lack of temporal continuity, class balance, and control-plane synchronisation often leads to irregular loss decay and gradient instability [12]. This improvement arises from the ASEADOS-SDN-IoT dataset's balanced class representation, coherent feature scaling, temporally aligned packet-flow sequences, and integrated control-plane telemetry. Together, these design choices create a continuous, learnable representation of network behaviour, promoting the formation of discriminative latent features and robust detection across varying attack contexts. Although this study evaluates the dataset using conventional ML and DL models at the controller level, the use of lightweight statistical features ensures compatibility with TinyML-based IDS implementations on resource-constrained IoT devices, which remains an important direction for future work.

Collectively, these findings confirm that ASEADOS-SDN-IoT bridges a critical gap in SDN-IoT intrusion detection research by unifying realistic packet-flow generation, temporal fidelity, and synchronised control-plane observability. Unlike previous datasets that isolate or emulate these aspects, ASEADOS-SDN-IoT provides a comprehensive foundation for ML and DL models to capture the full semantics of SDN-IoT interactions. This establishes it as a next-generation benchmark for developing adaptive, context-aware, and self-optimising intrusion detection frameworks in programmable IoT environments.

8. Future research directions

The ASEADOS-SDN-IoT dataset, together with its documented testbed framework, provides a solid foundation for advancing research on intelligent, cross-layer cybersecurity for programmable IoT environments. By combining synchronised control-plane telemetry, realistic packet-flow monitoring, and a hybrid physical-virtual design, the framework delivers comprehensive visibility into real SDN-IoT interactions. This integrated approach transforms ASEADOS-SDN-IoT from a conventional dataset into a complete experimental platform that supports reproducible testing, comparative model evaluation, and in-depth analysis of controller behaviour, attack dynamics, and defensive adaptation. In addition, it establishes a flexible and extensible foundation for long-term research and innovation in SDN-IoT security analytics and testbed design.

8.1. Scalable and evolving SDN-IoT research framework

A key future direction is to expand the ASEADOS-SDN-IoT testbed to a large-scale, heterogeneous environment that reflects the diversity and complexity of real-world IoT ecosystems. Incorporating ultra-constrained sensor nodes, industrial control systems, vehicular devices, and multi-protocol gateways using technologies such as 6LoWPAN, Zigbee, LoRaWAN, and 5G would enable deeper exploration of cross-domain communication patterns and inter-protocol interference. Such scaling would allow researchers to study emerging challenges in distributed orchestration, intent-based networking, and service-level isolation in high-density IoT environments. Moreover, scaling the framework to cloud-edge-fog hierarchies can facilitate evaluation of decentralised traffic management and resource offloading under realistic latency and throughput conditions. In parallel, the framework should evolve into a continuously updated research ecosystem through automated telemetry collection, online labelling, and adaptive sampling, ensuring that the dataset grows dynamically as network behaviours and threat patterns change over time. This integrated approach would transform ASEADOS-SDN-IoT into a sustainable, collaborative, and reproducible platform for long-term research and innovation in intelligent SDN-IoT security systems.

8.2. Distributed and collaborative SDN architectures

Another research approach can focus on extending the current single-controller topology to distributed and hierarchical SDN architectures. Multi-controller deployments can uncover complex behaviours related to state synchronisation, policy consistency, and dynamic load redistribution. Future experiments should leverage the ASEADOS-SDN-IoT framework to examine controller clustering, cooperative defence coordination, and consensus-based control under adversarial conditions. These studies could also support the development of self-organising and fault-tolerant control planes capable of maintaining stability, scalability, and attack resilience in SDN-IoT infrastructures.

8.3. Feature enrichment and encrypted traffic intelligence

Future iterations of the dataset should incorporate higher-layer (Application) and encrypted traffic features to enable privacy-aware yet effective intrusion detection. Integrating encrypted flow fingerprints, packet timing correlations, and statistical side-channel indicators would facilitate the design of privacy-preserving IDS models that operate without decryption. This direction bridges the gap between regulatory compliance and analytical capability, supporting research in federated intrusion detection, homomorphic encryption, and lightweight feature abstraction for constrained IoT environments. By doing so, future versions of ASEADOS-SDN-IoT can contribute to secure data analytics pipelines that maintain both confidentiality and model explainability.

8.4. Modelling adaptive and AI-driven threat ecosystems

The ASEADOS-SDN-IoT framework offers an opportunity to simulate adaptive, AI-driven, and multi-stage threat campaigns that mirror modern attack evolution. Future research can model reinforcement-learning-based adversaries that adapt their tactics in response to network defences, thereby evaluating IDS robustness under dynamic threat adaptation. Extending the attack spectrum to include controller poisoning, ransomware propagation, and supply-chain compromise would enable systematic assessment of detection models under intelligent adversarial conditions. This direction will promote the co-evolution of autonomous defensive agents and self-healing SDN controllers, advancing toward resilient cyber-physical defense ecosystems.

8.5. Intelligent and explainable detection paradigms

The synchronised telemetry available in ASEADOS-SDN-IoT provides a foundation for developing intelligent, interpretable IDS frameworks. Future studies should exploit graph neural networks, causal inference models, and attention-based architectures to uncover temporal-spatial dependencies between packet flows, controller decisions, and network state transitions. The integration of causal graph reasoning and explainable AI (XAI) techniques can enhance transparency, allowing operators to trace the decision pathways of IDS predictions. These approaches would not only improve detection accuracy but also enable actionable situational awareness, thereby increasing trust in AI-driven SDN-IoT defence systems.

In summary, the ASEADOS-SDN-IoT dataset and framework introduce an extensible research platform that bridges empirical experimentation with scalable SDN-IoT security innovation. The outlined directions-spanning large-scale testbed development, distributed control, encrypted data analytics, adaptive adversaries, and explainable intelligence open a roadmap for the next decade of SDN-IoT intrusion detection research. Together, these directions move the field toward autonomous, explainable, and adaptive network defence, establishing ASEADOS-SDN-IoT as a foundation for next-generation SDN-IoT cybersecurity research.

9. Conclusion

This study introduced ASEADOS-SDN-IoT, a novel and comprehensive dataset and testbed framework designed to advance intrusion detection research in programmable SDN-IoT environments. The work addresses one of the most persistent challenges in intelligent network security-the absence of realistic, integrated benchmarks that capture the interplay between SDN control intelligence and heterogeneous IoT traffic behaviour. Unlike existing datasets that isolate either SDN or IoT aspects, ASEADOS-SDN-IoT

integrates synchronised control-plane telemetry and data-plane packet captures within a hybrid experimental framework combining physical IoT devices, virtual nodes, and an ONOS-based SDN controller. This design ensures realistic modelling of flow behaviours, controller interactions, and attack patterns across multiple threat categories, including DoS, DDoS, Botnet, and Probe scenarios.

Extensive evaluation using both ML and DL techniques validated the dataset's discriminative power and representative quality. Ensemble-based classifiers and neural architectures achieved consistently high accuracy, recall, and F1-scores across all attack types, confirming ASEADOS-SDN-IoT's robustness, feature diversity, and suitability for adaptive IDS development. Beyond its empirical validation, the framework provides a reproducible foundation for future studies in cross-layer, privacy-aware, and context-driven intrusion detection. By integrating realism, scalability, and cross-domain observability, ASEADOS-SDN-IoT establishes not only a benchmark dataset but also a flexible research infrastructure that supports ongoing innovation in SDN-IoT cybersecurity. Ultimately, it represents a significant step toward developing resilient, explainable, and self-adaptive defence systems for next-generation programmable IoT networks.

Dataset availability

The ASEADOS-SDN-IoT dataset is publicly available on <https://aseados.ucd.ie/datasets/SDN-IoT/> with the publishing of this paper.

CRedit authorship contribution statement

Tharindu Lakshan Yasarathna: Writing - review & editing, Writing - original draft, Visualization, Validation, Methodology, Data curation; **Nhien-An Le-Khac:** Writing – review & editing, Supervision, Resources, Investigation.

Data availability

The dataset used in this study is accessible via the link provided in the Data Availability section of the manuscript.

Declaration of competing interest

Given their role as AE, Nhien-An Le-Khac had no involvement in the peer-review of this article and has no access to information regarding its peer-review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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